

Docker Certified Associate Exam

180 Real Exam Question and Answers



DOCKER
CERTIFIED ASSOCIATE

DCA



ShapingPixel

<https://shapingpixel.com>



PDF

1) Which of the following commands will automatically create a volume when a container is started?

A. 'docker container run --name nginxtest --volumes=/app nginx'

B. 'docker container run --name nginxtest -v /app:mount nginx'

C. 'docker container run --name nginxtest --volumes
myvol:/app:new nginx'

D. 'docker container run --name nginxtest -v myvol:/app nginx'

1) Which of the following commands will automatically create a volume when a container is started?

A. 'docker container run --name nginxtest --volumes=/app nginx'

B. 'docker container run --name nginxtest -v /app:mount nginx'

C. 'docker container run --name nginxtest --volumes
myvol:/app:new nginx'

D. 'docker container run --name nginxtest -v myvol:/app nginx'

Answer(s): A

2) Which one of the following commands will show a list of volumes for a specific container?

- A. 'docker container logs nginx--volumes'
- B. 'docker container inspect nginx'
- C. 'docker volume inspect nginx'
- D. 'docker volume logs nginx --containers'

2) Which one of the following commands will show a list of volumes for a specific container?

- A. 'docker container logs nginx--volumes'
- B. 'docker container inspect nginx'
- C. 'docker volume inspect nginx'
- D. 'docker volume logs nginx --containers'

Answer(s): B

3) Which of the following constitutes a production-ready device mapper configuration for the Docker engine?

- A. Create a volume group in device mapper and utilize the '--dm.thinpooldev' Docker daemon option, specifying the volume group
- B. Format a partition with xfs and mount it at '/var/lib/docker'
- C. Utilize the '--storage-opt dm.directlvm_device' Docker daemon option, specifying a block device
- D. Nothing, device mapper comes ready for production usage out of the box

3) Which of the following constitutes a production-ready device mapper configuration for the Docker engine?

- A. Create a volume group in device mapper and utilize the '--dm.thinpooldev' Docker daemon option, specifying the volume group
- B. Format a partition with xfs and mount it at '/var/lib/docker'
- C. Utilize the '--storage-opt dm.directlvm_device' Docker daemon option, specifying a block device
- D. Nothing, device mapper comes ready for production usage out of the box

Answer(s): C

4) Which one of the following commands will result in the volume being removed automatically once the container has exited?

- A. 'docker run --del -v /foobusybox'
- B. 'docker run --read-only -v /foobusybox'
- C. 'docker run --rm -v /foobusybox'
- D. 'docker run --remove -v /foobusybox'

4) Which one of the following commands will result in the volume being removed automatically once the container has exited?

- A. 'docker run --del -v /foobusybox'
- B. 'docker run --read-only -v /foobusybox'
- C. 'docker run --rm -v /foobusybox'
- D. 'docker run --remove -v /foobusybox'

Answer(s): C

5) A container named "analytics" that stores results in a volume called "data" was created. `docker run -d -name=analytics -v data:/data app1`

How are the results accessed in "data" with another container called "app2"?

A. `docker run -d --name=reports --volume=data app2`

B. `docker run -d --name=reports --volumes-from=analytics app2`

C. `docker run -d --name=reports --volume=app1 app2`

D. `docker run -d --name=reports --mount=app1 app2`

5) A container named "analytics" that stores results in a volume called "data" was created. `docker run -d -name=analytics -v data:/data app1`

How are the results accessed in "data" with another container called "app2"?

A. `docker run -d --name=reports --volume=data app2`

B. `docker run -d --name=reports --volumes-from=analytics app2`

C. `docker run -d --name=reports --volume=app1 app2`

D. `docker run -d --name=reports --mount=app1 app2`

Answer(s): B

6)A server is running low on disk space. What command can be used to check the disk usage of images, containers, and volumes for Dockerengine?

- A.'docker system df'
- B.'docker system prune'
- C.'docker system free'
- D.'docker system ps'

6)A server is running low on disk space. What command can be used to check the disk usage of images, containers, and volumes for Dockerengine?

- A.'docker system df'
- B.'docker system prune'
- C.'docker system free'
- D.'docker system ps'

Answer(s): A

7) Which of the following are types of namespaces used by Docker to provide isolation? (Choose 2.)

A. Host

B. Network

C. Process ID

D. Authentication

E. Storage

7) Which of the following are types of namespaces used by Docker to provide isolation? (Choose 2.)

- A. Host
- B. Network
- C. Process ID
- D. Authentication
- E. Storage

Answer(s): B,C

8) Which of the following namespaces is disabled by default and must be enabled at Docker engine runtime in order to be used?

A. user

B. pid

C. net

D. mnt

8) Which of the following namespaces is disabled by default and must be enabled at Docker engine runtime in order to be used?

A. user

B. pid

C. net

D. mnt

Answer(s): A

9) Which of the following statements is true about secrets?

A. Secrets can be created from any node in the cluster.

B. Secrets can be modified after they are created.

C. Secret are stored unencrypted on manager nodes.

D. Secrets can be created using standard input (STDIN) and a file.

9) Which of the following statements is true about secrets?

A. Secrets can be created from any node in the cluster.

B. Secrets can be modified after they are created.

C. Secret are stored unencrypted on manager nodes.

D. Secrets can be created using standard input (STDIN) and a file.

Answer(s): D

10)Following the principle of least privilege, which of the following methods can be used to securely grant access to the specific user to communicate to a Docker engine? (Choose two.)

- A.Utilize the '--host 0.0.0.0:2375' option to the Docker daemon to listen on port 2375 over TCP on all interfaces
- B.Utilize open ssl to create TLSclient and server certificates, configuring the Docker engine to use with mutual TLS over TCP.
- C.Utilize the '--host 127.0.0.1:2375' option to the Docker daemon to listen on port 2375 over TCP on localhost
- D.Give the user root access to the server to allow them to run Docker commands as root.
- E.Add the user to the 'docker' group on the server or specify the group with the '--group' Docker daemon option.

10)Following the principle of least privilege, which of the following methods can be used to securely grant access to the specific user to communicate to a Docker engine? (Choose two.)

- A.Utilize the '--host 0.0.0.0:2375' option to the Docker daemon to listen on port 2375 over TCP on all interfaces
- B.Utilize open ssl to create TLSclient and server certificates, configuring the Docker engine to use with mutual TLS over TCP.
- C.Utilize the '--host 127.0.0.1:2375' option to the Docker daemon to listen on port 2375 over TCP on localhost
- D.Give the user root access to the server to allow them to run Docker commands as root.
- E.Add the user to the 'docker' group on the server or specify the group with the '--group' Docker daemon option.

Answer(s): B,E

11) Which of the following is supported by control groups?

- A. Manage certificates
- B. Collect net
- C. Limit CPU usage within a container
- D. Isolate processes in a container

11) Which of the following is supported by control groups?

- A. Manage certificates
- B. Collect net
- C. Limit CPU usage within a container
- D. Isolate processes in a container

Answer(s): C

12)What is the purpose of Docker Content Trust?

- A.Signing and verification of imagetags
- B.Enabling mutual TLS between the Docker client and server
- C.Docker registry TLS verification and encryption
- D.Indicating an image on Docker Hub is an official image

12)What is the purpose of Docker Content Trust?

- A.Signing and verification of imagetags
- B.Enabling mutual TLS between the Docker client and server
- C.Docker registry TLS verification and encryption
- D.Indicating an image on Docker Hub is an official image

Answer(s): A

13)What is the purpose of a client bundle in the Universal Control Plane?

- A.Authenticate a user using client certificates to the Universal Control Plane
- B.Provide a new user instructions for how to login to the Universal Control Plane
- C.Provide a user with a Docker client binary compatible with the Universal Control Plane
- D.Group multiple users in a team in the Universal Control Plane

13)What is the purpose of a client bundle in the Universal Control Plane?

- A.Authenticate a user using client certificates to the Universal Control Plane
- B.Provide a new user instructions for how to login to the Universal Control Plane
- C.Provide a user with a Docker client binary compatible with the Universal Control Plane
- D.Group multiple users in a team in the Universal Control Plane

Answer(s): A

14)When using the Docker client to push an image to a registry, what environment variable is used to instruct the client to perform signing of the image?

A.DOCKER_CONTENT_TRUST=1

B.DOCKER_IMAGE_SIGN=1

C.DOCKER_PUSH_SIGN=1

D.NOTARY_ENABLE=1 Correct

14)When using the Docker client to push an image to a registry, what environment variable is used to instruct the client to perform signing of the image?

A.DOCKER_CONTENT_TRUST=1

B.DOCKER_IMAGE_SIGN=1

C.DOCKER_PUSH_SIGN=1

D.NOTARY_ENABLE=1 Correct

Answer(s): A

15) You have created a Docker bridge network on a host with three containers attached, how do you make these containers accessible outside of the host?

A. Use network attach to access the containers on the bridge network

B. Use either EXPOSE or --publish to access the containers on the bridge network

C. Use network connect to access the containers on the bridge network

D. Use --link to access the containers on the bridge network

Correct

15) You have created a Dockerbridge network on a host with three containers attached, how do you make these containers accessible outside of the host?

A. Use network attach to access the containers on the bridge network

B. Use either EXPOSE or --publish to access the containers on the bridge network

C. Use network connect to access the containers on the bridge network

D. Use --link to access the containers on the bridge network
Correct

Answer(s): B

16) Which of the following commands will ensure that overlay traffic between service tasks is encrypted?

A. `docker service create --network <network-name> --secure <service-name>`

B. `docker network create -d overlay --secure <network-name>`

C. `docker network create -d overlay -o encrypted=true <network-name>`

D. `docker service create --network <network-name> --encrypted <service-name>`

16) Which of the following commands will ensure that overlay traffic between service tasks is encrypted?

A. `docker service create --network <network-name> --secure <service-name>`

B. `docker network create -d overlay --secure <network-name>`

C. `docker network create -d overlay -o encrypted=true <network-name>`

D. `docker service create --network <network-name> --encrypted <service-name>`

Answer(s): C

17) Which of the following commands will create a swarm service which only listens on port 53 using the UDP protocol?

A. `docker service create --name dns-cache -p 53:53/udp dns-cache`

B. `docker service create --name dns-cache -p 53:53 --service udp dns-cache`

C. `docker service create --name dns-cache -p 53:53 --constraint networking.protocol.udp=true dns-cache`

D. `docker service create --name dns-cache -p 53:53 --udp dns-cache`

17) Which of the following commands will create a swarm service which only listens on port 53 using the UDP protocol?

A. `docker service create --name dns-cache -p 53:53/udp dns-cache`

B. `docker service create --name dns-cache -p 53:53 --service udp dns-cache`

C. `docker service create --name dns-cache -p 53:53 --constraint networking.protocol.udp=true dns-cache`

D. `docker service create --name dns-cache -p 53:53 --udp dns-cache`

Answer(s): A

18) Which of the following is true about using the '-P' option when creating a new container?

- A. Docker binds each exposed container port to a random port on all the host's interface
- B. Docker gives extended privileges to the container.
- C. Docker binds each exposed container port to a random port on a specified host interface
- D. Docker binds each exposed container port with the same port on the host

18) Which of the following is true about using the '-P' option when creating a new container?

- A. Docker binds each exposed container port to a random port on all the host's interface
- B. Docker gives extended privileges to the container.
- C. Docker binds each exposed container port to a random port on a specified host interface
- D. Docker binds each exposed container port with the same port on the host

Answer(s): A

19)What behavior is expected when a service is created with the following command: 'docker service create --publish 8000:80 nginx'

- A.All nodes in the cluster will listen on port 8080 and forward to port 80 in the container.
- B.Only a single node in the cluster will listen on port 8080 and forward to port 80 in the container.
- C.All nodes in the cluster will listen on port 80 and forward to port 8080 in the container.
- D.Only a single node in the cluster will listen on port 80 and forward to port 8080 in the container.

19)What behavior is expected when a service is created with the following command: 'docker service create --publish 8000:80 nginx'

A.All nodes in the cluster will listen on port 8080 and forward to port 80 in the container.

B.Only a single node in the cluster will listen on port 8080 and forward to port 80 in the container.

C.All nodes in the cluster will listen on port 80 and forward to port 8080 in the container.

D.Only a single node in the cluster will listen on port 80 and forward to port 8080 in the container.

Answer(s): A

20) Which set of commands can identify the published port(s) for a container? (Choose 1.)

- A. 'docker port inspect', 'docker container inspect'
- B. 'docker container inspect', 'dockerport'
- C. 'docker info', 'docker network inspect'
- D. 'docker network inspect', 'docker port'

20) Which set of commands can identify the published port(s) for a container? (Choose 1.)

- A. 'docker port inspect', 'docker container inspect'
- B. 'docker container inspect', 'docker port'
- C. 'docker info', 'docker network inspect'
- D. 'docker network inspect', 'docker port'

Answer(s): B

21) Which of the following is true about overlay networks?

- A. Overlay networks are created only on the manager node that you created the overlay networking on
- B. Overlay networks are created on all cluster nodes when you create the overlay network.
- C. Overlay networks are first created on the manager nodes. Then they are created on the worker nodes once a task is scheduled on the specific worker node.
- D. Overlay networks are only created on the manager nodes.

21) Which of the following is true about overlay networks?

- A. Overlay networks are created only on the manager node that you created the overlay networking on
- B. Overlay networks are created on all cluster nodes when you create the overlay network.
- C. Overlay networks are first created on the manager nodes. Then they are created on the worker nodes once a task is scheduled on the specific worker node.
- D. Overlay networks are only created on the manager nodes.

Answer(s): B

22) Which of the following mode can be used for service discovery of a Docker Swarm service (Pick 2 correct answers)

- A. Virtual IP (VIP) with `--endpoint-mode vip`
- B. Overlay with `--endpoint-mode overlay`
- C. DNS Round-Robin with `--endpoint-mode dnsrr`
- D. Ingress with `--endpoint-mode ingress`
- E. Network Address Translation (NAT) with `--endpoint-mode nat`

22) Which of the following mode can be used for service discovery of a Docker Swarm service (Pick 2 correct answers)

- A. Virtual IP (VIP) with `--endpoint-mode vip`
- B. Overlay with `--endpoint-mode overlay`
- C. DNS Round-Robin with `--endpoint-mode dnsrr`
- D. Ingress with `--endpoint-mode ingress`
- E. Network Address Translation (NAT) with `--endpoint-mode nat`

Answer(s): A,C

23) A user is having problems running Docker. Which of the following will start Docker in debug mode?

- A. Set the debug key to true in the 'daemon.json' file.
- B. Start the 'dockerd' process manually with the '--logging' flag set to debug
- C. Set the logging key to debug in the 'daemon.json' file.
- D. Start the 'dockerd' process manually with the '--raw-logs' flag set to debug

23) A user is having problems running Docker. Which of the following will start Docker in debug mode?

- A. Set the debug key to true in the 'daemon.json' file.
- B. Start the 'dockerd' process manually with the '--logging' flag set to debug
- C. Set the logging key to debug in the 'daemon.json' file.
- D. Start the 'dockerd' process manually with the '--raw-logs' flag set to debug

Answer(s): A

24) Which command interactively monitors all container activity in the Docker engine?

- A. docker system logs
- B. docker system events
- C. docker container events
- D. docker container logs

24) Which command interactively monitors all container activity in the Docker engine?

- A. docker system logs
- B. docker system events
- C. docker container events
- D. docker container logs

Answer(s): B

25)What is used by the kernel to Isolate resources when running Docker containers?

- A.Namespaces
- B.Overlay networks
- C.Volumes
- D.Control groups (also know as cgroups)

25)What is used by the kernel to Isolate resources when running Docker containers?

- A.Namespaces
- B.Overlay networks
- C.Volumes
- D.Control groups (also know as cgroups)

Answer(s): D

26) A host machine has four CPUs available and two running containers. The sysadmin would like to assign two CPUs to each container.

Which of the following commands achieves this?

- A. Set the '--cpuset-cpu's flag to '1,3' on one container and '2,4' on the other container.
- B. Set the '--cpuset-cpus' flag to '.5' on both containers
- C. Set the '--cpuset-cpus' flag of the 'dockerd' process to the value 'even-spread'
- D. Set the '--cpu-quota' flag to '1,3' on one container and '2,4' on the other container.

26) A host machine has four CPUs available and two running containers. The sysadmin would like to assign two CPUs to each container.

Which of the following commands achieves this?

- A. Set the '--cpuset-cpu's flag to '1,3' on one container and '2,4' on the other container.
- B. Set the '--cpuset-cpus' flag to '.5' on both containers
- C. Set the '--cpuset-cpus' flag of the 'dockerd' process to the value 'even-spread'
- D. Set the '--cpu-quota' flag to '1,3' on one container and '2,4' on the other container.

Answer(s): B

27) Which of the following commands is used to display system-wide Docker configuration on a host?

- A. docker info
- B. docker status
- C. docker inspect
- D. docker system

27) Which of the following commands is used to display system-wide Docker configuration on a host?

- A. docker info
- B. docker status
- C. docker inspect
- D. docker system

Answer(s): A

28) If installing Docker using device mapper for storage with the intent to run production work loads, how should device mapper be configured

- A. direct-lvm
- B. loop-lvm
- C. overlay-lvm
- D. aufs-lvm

28) If installing Docker using device mapper for storage with the intent to run production work loads, how should device mapper be configured

- A. direct-lvm
- B. loop-lvm
- C. overlay-lvm
- D. aufs-lvm

Answer(s): A

29) Which of the following is required to install Docker EE from a package repository?

- A. Repository URL obtained from Docker Store
- B. License key obtained from DockerStore
- C. Repository URL obtained from DockerHub
- D. License key obtained from Docker Hub

29) Which of the following is required to install Docker EE from a package repository?

- A. Repository URL obtained from Docker Store
- B. License key obtained from DockerStore
- C. Repository URL obtained from DockerHub
- D. License key obtained from Docker Hub

Answer(s): A

30)How do you configure Docker engine to use a registry that is not configured with TLS certificates from a trusted CA?

- A.Set IGNORE_TLS in the 'daemon.json' configuration file.
- B.Set and export the IGNORE_TLS environment variable on the command line
- C.Set INSECURE_REGISTRY in the '/etc/docker/default' configuration file
- D.Pass the '--insecure.-registry' flag to the daemon at run time

30)How do you configure Docker engine to use a registry that is not configured with TLS certificates from a trusted CA?

- A.Set IGNORE_TLS in the 'daemon.json' configuration file.
- B.Set and export the IGNORE_TLS environment variable on the command line
- C.Set INSECURE_REGISTRY in the '/etc/docker/default' configuration file
- D.Pass the '--insecure.-registry' flag to the daemon at run time

Answer(s): D

31)The output of which command can be used to find the architecture and operating system an image is compatible with?

- A.docker image inspect --filter {{.Architecture}} {{.OS}} ' <image-id>
- B.docker image ls <image-id>
- C.docker image inspect --format {{.Architecture}} {{.OS}} ' <image-id>
- D.docker image info <image-id>

31)The output of which command can be used to find the architecture and operating system an image is compatible with?

- A.docker image inspect --filter {{.Architecture}} {{.OS}} ' <image-id>
- B.docker image ls <image-id>
- C.docker image inspect --format {{.Architecture}} {{.OS}} ' <image-id>
- D.docker image info <image-id>

Answer(s): C

32)An application image runs in multiple environments, and each environment uses different certificates and ports, what is the best practice to deploy the containers?

- A.Create a Dockerfile for each environment, specifying ports and ENV variables for certificates.
- B.Create a Dockerfile for each environment, specifying ports and Docker secrets for certificates.
- C.Create images that contain the specific configuration for every environment.
- D.Create a config file for each environment.

32)An application image runs in multiple environments, and each environment uses different certificates and ports, what is the best practice to deploy the containers?

- A.Create a Dockerfile for each environment, specifying ports and ENV variables for certificates.
- B.Create a Dockerfile for each environment, specifying ports and Docker secrets for certificates.
- C.Create images that contain the specific configuration for every environment.
- D.Create a config file for each environment.

Answer(s): D

33)What is the purpose of multi-stage builds?

- A.Better logical separation of Docker file instructions for better readability
- B.Optimizing images by copying artifacts selectively from previous stages
- C.Better caching when building Docker images
- D.Faster image builds by allowing parallel execution of Docker builds

33)What is the purpose of multi-stage builds?

- A.Better logical separation of Docker file instructions for better readability
- B.Optimizing images by copying artifacts selectively from previous stages
- C.Better caching when building Docker images
- D.Faster image builds by allowing parallel execution of Docker builds

Answer(s): B

34) From a DevOps process standpoint, it is best practice to keep changes to an application in version control. Which of the following will allow changes to a docker Image to be stored in a version control system?

- A. docker commit
- B. docker save
- C. A docker-compose.yml file
- D. A dockerfile

34) From a DevOps process standpoint, it is best practice to keep changes to an application in version control. Which of the following will allow changes to a docker Image to be stored in a version control system?

- A. docker commit
- B. docker save
- C. A docker-compose.yml file
- D. A dockerfile

Answer(s): A

35) When seven managers are in a swarm cluster how would they be distributed across three datacenters or availability zones?

A. 5-1-1

B. 3-2-2

C. 3-3-1

D. 4-2-1

35)When seven managers are in a swarm cluster how would they be distributed across three datacenters or availability zones?

A.5-1-1

B.3-2-2

C.3-3-1

D.4-2-1

Answer(s): B

36)What is the difference between a resource limit and a resource reservation when scheduling services?

A.A resource limit and a resource reservation can be used interchangeably.

B.A resource limit is a soft limit for your service, while a reservation is hard limit and the docker engine will do its best to keep your service at the limit.

C.A resource limit is used to find a host with adequate resources for scheduling a hard limit for your service, while a reservation is hard limit for your service.

D.A resource limit is hard limit for your service, while a reservation is used to find a host with adequate resources for scheduling. Correct

36)What is the difference between a resource limit and a resource reservation when scheduling services?

A.A resource limit and a resource reservation can be used interchangeably.

B.A resource limit is a soft limit for your service, while a reservation is hard limit and the docker engine will do its best to keep your service at the limit.

C.A resource limit is used to find a host with adequate resources for scheduling a hard limit for your service, while a reservation is hard limit for your service.

D.A resource limit is hard limit for your service, while a reservation is used to find a host with adequate resources for scheduling. Correct

Answer(s): A

37) You have just executed 'docker swarm leave' on a node. What command can be run on the same node to confirm it has left the cluster?

- A. docker node ls
- B. docker system info
- C. docker system status
- D. docker system status

37) You have just executed 'docker swarm leave' on a node. What command can be run on the same node to confirm it has left the cluster?

- A. docker node ls
- B. docker system info
- C. docker system status
- D. docker system status

Answer(s): B

38)What is the recommended way to configure the daemon flags and environment variables for your Docker daemon in a platform independent way?

- A.Set the configuration options using the ENV variable
- B.Set the configuration options in '/etc/docker/daemon.json'
- C.Set the configuration DOCKER_OPTS in '/etc/default/docker'
- D.Using 'docker config' to set the configuration options.

38)What is the recommended way to configure the daemon flags and environment variables for your Docker daemon in a platform independent way?

- A.Set the configuration options using the ENV variable
- B.Set the configuration options in '/etc/docker/daemon.json'
- C.Set the configuration DOCKER_OPTS in '/etc/default/docker'
- D.Using 'docker config' to set the configuration options.

Answer(s): B

39) You have deployed a service to swarm. Which command uses the Docker CLI to set the number of tasks of the services to 5? (Choose 2)

- A. 'docker service update --replicas=5<service-id>'
- B. 'docker replica update <service-id>=5'
- C. 'docker update service <service-id>=5'
- D. 'docker service replicas <service-id>=5'
- E. 'docker service scale <service-id> =5"

39) You have deployed a service to swarm. Which command uses the Docker CLI to set the number of tasks of the services to 5? (Choose 2)

- A. 'docker service update --replicas=5<service-id>'
- B. 'docker replica update <service-id>=5'
- C. 'docker update service <service-id>=5'
- D. 'docker service replicas <service-id>=5'
- E. 'docker service scale <service-id> =5"

Answer(s): A,E

40) A service 'wordpress' is running using a password string to connect to a non-Dockerized database service. The password string is passed into the 'wordpress' service as a Docker secret. Per security policy, the password on the database was changed. Identity the correct sequence of steps to rotate the secret from the old password to the new password.

- A. Create a new docker secret with the new password. Trigger a rolling secret update by using the 'docker secret update' command
- B. Trigger an update to the service by using 'docker service update --secret=<new password>'
- C. Create a new docker secret with the new password. Remove the existing service using 'docker service rm'. Start a new service with the new secret using "--secret=<new password>"
- D. Create a new docker secret with a new password. Trigger a rolling update of the "wordpress" service, by using "--secret-rm" & "--secret-add" to remove the old secret and add the updated secret.

40) A service 'wordpress' is running using a password string to connect to a non-Dockerized database service. The password string is passed into the 'wordpress' service as a Docker secret. Per security policy, the password on the database was changed. Identify the correct sequence of steps to rotate the secret from the old password to the new password.

- A. Create a new docker secret with the new password. Trigger a rolling secret update by using the 'docker secret update' command
- B. Trigger an update to the service by using 'docker service update --secret=<new password>'
- C. Create a new docker secret with the new password. Remove the existing service using 'docker service rm'. Start a new service with the new secret using "--secret=<new password>"
- D. Create a new docker secret with a new password. Trigger a rolling update of the "wordpress" service, by using "--secret-rm" & "--secret-add" to remove the old secret and add the updated secret.

Answer(s): D

41)What service mode is used to deploy a single task of a service to each node?

- A.replicated
- B.spread
- C.universal
- D.distributed
- E.global

41)What service mode is used to deploy a single task of a service to each node?

- A.replicated
- B.spread
- C.universal
- D.distributed
- E.global

Answer(s): E

42) A docker service 'web' is running with a scale factor of 1 (replicas = 1). Bob intends to use the command 'docker service update --replicas=3 web'. Alice intends to use the command 'docker service scale web=3'.

How do the outcomes of these two commands differ?

- A. Bob's command results in an error. Alice's command updates the number of replicas of the 'web' service to 3.
- B. Bob's command only updates the service definition, but no new replicas are started. Alice's command results in the actual scaling up of the 'web' service.
- C. Bob's command updates the number of replicas of the 'web' service to 3. Alice's command results in an error.
- D. Both Bob's and Alice's commands result in exactly the same outcome, which is 3 instances of the 'web' service.

42) A docker service 'web' is running with a scale factor of 1 (replicas = 1). Bob intends to use the command 'docker service update --replicas=3 web'. Alice intends to use the command 'docker service scale web=3'.

How do the outcomes of these two commands differ?

- A. Bob's command results in an error. Alice's command updates the number of replicas of the 'web' service to 3.
- B. Bob's command only updates the service definition, but no new replicas are started. Alice's command results in the actual scaling up of the 'web' service.
- C. Bob's command updates the number of replicas of the 'web' service to 3. Alice's command results in an error.
- D. Both Bob's and Alice's commands result in exactly the same outcome, which is 3 instances of the 'web' service.

Answer(s): D

43)The following health check exists in a Dockerfile:

'HEALTHCHECKCMDcurl --fail http://localhost/health ||exit 1' Which of the following describes its purpose?

- A.Defines the action taken when container health fails, which in this case will kill the container with exit status 1
- B.Defines the health check endpoint on the localhost interface for external monitoring tools to monitor the health of the docker engine.
- C.Defines the health check endpoint on the local host interface for containers to monitor the health of the docker engine.
- D.Defines the health check for the containerized application so that the application health can be monitored by the Dockerengine

43)The following health check exists in a Dockerfile:

'HEALTHCHECKCMDcurl --fail http://localhost/health ||exit 1' Which of the following describes its purpose?

- A.Defines the action taken when container health fails, which in this case will kill the container with exit status 1
- B.Defines the health check endpoint on the localhost interface for external monitoring tools to monitor the health of the docker engine.
- C.Defines the health check endpoint on the local host interface for containers to monitor the health of the docker engine.
- D.Defines the health check for the containerized application so that the application health can be monitored by the Dockerengine

Answer(s): A

44) Which of these swarm manager configurations will cause the cluster to be in a lost quorum state?

- A. 4 managers of which 2 are healthy
- B. 1 manager of which 1 is healthy
- C. 3 managers of which 2 are healthy
- D. 5 managers of which 3 are healthy

44) Which of these swarm manager configurations will cause the cluster to be in a lost quorum state?

- A. 4 managers of which 2 are healthy
- B. 1 manager of which 1 is healthy
- C. 3 managers of which 2 are healthy
- D. 5 managers of which 3 are healthy

Answer(s): A

45) Which of the following commands starts a Redis container and configures it to always restart unless it is explicitly stopped or Docker is restarted?

- A. 'docker run -d --restart-policy unless-stopped redis'
- B. 'docker run -d --restart omit-stopped redis'
- C. 'docker run -d --restart unless-stopped redis'
- D. 'docker run -d --failure omit-stopped redis'

45) Which of the following commands starts a Redis container and configures it to always restart unless it is explicitly stopped or Docker is restarted?

- A. 'docker run -d --restart-policy unless-stopped redis'
- B. 'docker run -d --restart omit-stopped redis'
- C. 'docker run -d --restart unless-stopped redis'
- D. 'docker run -d --failure omit-stopped redis'

Answer(s): C

46)After creating a new service named 'http', you notice that the new service is not registering as healthy. How do you view the list of historical tasks for that service by using the command line?

- A.'docker inspect http'
- B.'docker service inspect http'
- C.'docker service ps http'
- D.'docker ps http'

46)After creating a new service named 'http', you notice that the new service is not registering as healthy. How do you view the list of historical tasks for that service by using the command line?

- A.'docker inspect http'
- B.'docker service inspect http'
- C.'docker service ps http'
- D.'docker ps http'

Answer(s): C

47) Which flag for a service would allow a container to consume more than 2 GB of memory only when there is no memory contention but would also prevent a container from consuming more than 4GB of memory, in any case?

- A. `--limit-memory 2GB --reserve-memory 4GB`
- B. `--limit-memory 4GB --reserve-memory 2GB`
- C. `--memory-swap 2GB --limit-memory 4GB`
- D. `--memory-swap 4GB --limit-memory 2GB`

47) Which flag for a service would allow a container to consume more than 2 GB of memory only when there is no memory contention but would also prevent a container from consuming more than 4GB of memory, in any case?

- A. `--limit-memory 2GB --reserve-memory 4GB`
- B. `--limit-memory 4GB --reserve-memory 2GB`
- C. `--memory-swap 2GB --limit-memory 4GB`
- D. `--memory-swap 4GB --limit-memory 2GB`

Answer(s): C

48) In Docker Trusted Registry, how would a user prevent an image, for example 'nginx:latest' from being overwritten by another user with push access to the repository?

- A. Tag the image with 'nginx:immutable'
- B. Remove push access from all other users.
- C. Use the DTR web UI to make the tag immutable.
- D. Keep a backup copy of the image on another repository.

48)In Docker Trusted Registry, how would a user prevent an image, for example 'nginx:latest' from being overwritten by another user with push access to the repository?

- A.Tag the image with 'nginx:immutable'
- B.Remove push access from all other users.
- C.Use the DTR web UI to make the tag immutable.
- D.Keep a backup copy of the image on another repository.

Answer(s): C

49) Which of the following is NOT backed up when performing a Docker Trusted backup operation?

- A. Access control to repos and images
- B. Repository metadata
- C. Image blobs
- D. DTR configurations

49) Which of the following is NOT backed up when performing a Docker Trusted backup operation?

- A. Access control to repos and images
- B. Repository metadata
- C. Image blobs
- D. DTR configurations

Answer(s): C

50) Which statement is true?

A. CMD shell format uses this form ["param", param, "param"]

B. ENTRYPOINT cannot be used in conjunction with CMD

C. CMD is used to run the software in the image along with any arguments

D. ENTRYPOINT cannot be overridden in the "docker container run" command

50) Which statement is true?

A. CMD shell format uses this form ["param", param, "param"]

B. ENTRYPOINT cannot be used in conjunction with CMD

C. CMD is used to run the software in the image along with any arguments

D. ENTRYPOINT cannot be overridden in the "docker container run" command

Answer(s): A

51)What is one way of directly transferring a Docker Image from one Docker host in another?

A.'docker push' the image to the IP address of the target host.

B.'docker commit' to save the image outside of the Docker filesystem. Then transfer the file over to the target host and 'docker start' to start the container again.

C.There is no way of directly transferring Docker images between hosts. A Docker Registry must be used as an intermediary.

D.'docker save' the image to save it as TAR file and copy it over to the target host. Then use 'docker load' to un-TAR the image back as a Docker image.

51)What is one way of directly transferring a Docker Image from one Docker host in another?

A.'docker push' the image to the IP address of the target host.

B.'docker commit' to save the image outside of the Docker filesystem. Then transfer the file over to the target host and 'docker start' to start the container again.

C.There is no way of directly transferring Docker images between hosts. A Docker Registry must be used as an intermediary.

D.'docker save' the image to save it as TAR file and copy it over to the target host. Then use 'docker load' to un-TAR the image back as a Docker image.

Answer(s): D

52) Which statement is true about DTR garbage collection?

- A. Garbage collection removes unreferenced image layers from DTR's backend storage.
- B. Garbage collection removes exited containers from cluster nodes.
- C. Garbage collection removes DTR images that are older than a configurable number of days.
- D. Garbage collection removes unused volumes from cluster nodes.

52) Which statement is true about DTR garbage collection?

- A. Garbage collection removes unreferenced image layers from DTR's backend storage.
- B. Garbage collection removes exited containers from cluster nodes.
- C. Garbage collection removes DTR images that are older than a configurable number of days.
- D. Garbage collection removes unused volumes from cluster nodes.

Answer(s): A

53)What is the difference between the ADD and COPY dockerfile instructions? (chosen 2)

- A.ADD supports compression format handling while COPY does not.
- B.COPY supports regular expression handling while ADD does not.
- C.COPY supports compression format handling while ADD does not.
- D.ADD support remote URL handling while COPY does not.
- E.ADD supports regular expression handling while COPY does not.

53)What is the difference between the ADD and COPY dockerfile instructions? (chosen 2)

- A.ADD supports compression format handling while COPY does not.
- B.COPY supports regular expression handling while ADD does not.
- C.COPY supports compression format handling while ADD does not.
- D.ADD support remote URL handling while COPY does not.
- E.ADD supports regular expression handling while COPY does not.

Answer(s): D,E

54)What is the docker command to find the current logging driver for a running container?

- A.docker stats
- B.docker info
- C.docker config
- D.docker inspect

54)What is the docker command to find the current logging driver for a running container?

- A.docker stats
- B.docker info
- C.docker config
- D.docker inspect

Answer(s): D

55)What is the docker command to setup a swarm?

- A.docker swarm init
- B.docker swarm create
- C.docker init swarm
- D.docker create swarm

55)What is the docker command to setup a swarm?

- A.docker swarm init
- B.docker swarm create
- C.docker init swarm
- D.docker create swarm

Answer(s): A

Explanation:

<https://docs.docker.com/engine/reference/commandline/swarm/>

56)Is this a supported user authentication method for Universal Control Plane?

Solution: PAM

A.Yes

B.No

56)Is this a supported user authentication method for Universal Control Plane? Solution: PAM

A.Yes

B.No

Answer(s): B

57) Will this sequence of steps completely delete an image from disk in the Docker Trusted Registry?

Solution: Delete the image and delete the image repository from Docker Trusted Registry

A. Yes

B. No

57) Will this sequence of steps completely delete an image from disk in the Docker Trusted Registry?

Solution: Delete the image and delete the image repository from Docker Trusted Registry

A. Yes

B. No

Answer(s): A

58) Will this sequence of steps completely delete an image from disk in the Docker Trusted Registry?

Solution: Delete the image and run garbage collection on the Docker Trusted Registry.

A. Yes

B. No

58) Will this sequence of steps completely delete an image from disk in the Docker Trusted Registry?

Solution: Delete the image and run garbage collection on the Docker Trusted Registry.

A. Yes

B. No

Answer(s): B

59)Is this the purpose of Docker Content Trust? Solution: Enable mutual TLS between the Docker client and server.

A.Yes

B.No

59)Is this the purpose of Docker Content Trust? Solution: Enable mutual TLS between the Docker client and server.

A.Yes

B.No

Answer(s): B

60)Is this the purpose of Docker Content Trust? Solution: Verify and encrypt Docker registry TLS.

A.Yes

B.No

60)Is this the purpose of Docker Content Trust? Solution: Verify and encrypt Docker registry TLS.

A.Yes

B.No

Answer(s): A

61)Is this a Linux kernel namespace that is disabled by default and must be enabled at Dockerengine runtime to be used?

Solution: mnt

A.Yes

B.No

61)Is this a Linux kernel namespace that is disabled by default and must be enabled at Dockerengine runtime to be used?

Solution: mnt

A.Yes

B.No

Answer(s): B

62)Is this a Linux kernel namespace that is disabled by default and must be enabled at Dockerengine runtime to be used?

Solution: net

A.Yes

B.No

62) Is this a Linux kernel namespace that is disabled by default and must be enabled at Docker engine runtime to be used?

Solution: net

A. Yes

B. No

Answer(s): B

63)Is this a Linux kernel namespace that is disabled by default and must be enabled at Dockerengine runtime to be used?

Solution: user

A.Yes

B.No

63)Is this a Linux kernel namespace that is disabled by default and must be enabled at Dockerengine runtime to be used?

Solution: user

A.Yes

B.No

Answer(s): A

64)Is this a way to configure the Docker engine to use a registry without a trusted TLS certificate? Solution:
Pass the '--insecure-registry' flag to the daemon at run time.

A.Yes

B.No

64)Is this a way to configure the Docker engine to use a registry without a trusted TLS certificate? Solution:
Pass the '--insecure-registry' flag to the daemon at run time.

A.Yes

B.No

Answer(s): A

65)The Kubernetes yaml shown below describes a networkPolicy.

Will the networkPolicy BLOCK this traffic?

Solution: a request issued from a pod bearing the tier: backend label, to a pod bearing the tier: frontend label

```
---yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: dca
  namespace: default
spec:
  podSelector:
    matchLabels:
      tier: backend
  ingress:
    - from:
      - podSelector:
          matchLabels:
            tier: api
    ...
```

A.Yes

B.No

65)The Kubernetes yaml shown below describes a networkPolicy.

Will the networkPolicy BLOCK this traffic?

Solution: a request issued from a pod bearing the tier: backend label, to a pod bearing the tier: frontend label

```
---yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: dca
  namespace: default
spec:
  podSelector:
    matchLabels:
      tier: backend
  ingress:
    - from:
      - podSelector:
          matchLabels:
            tier: api
    ...
```

A.Yes

B.No

Answer(s): B

66)The Kubernetes yaml shown below describes a networkPolicy.

Will the networkPolicy BLOCK this traffic?

Solution: are quest issued from a pod lacking the tier: api label, to a pod bearing the tier: backend label

```
---yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: dca
  namespace: default
spec:
  podSelector:
    matchLabels:
      tier: backend
  ingress:
    - from:
      - podSelector:
          matchLabels:
            tier: api
    ...
```

A.Yes

B.No

66)The Kubernetes yamI shown below describes a networkPolicy.

Will the networkPolicy BLOCK this traffic?

Solution: are quest issued from a pod lacking the tier: api label, to a pod bearing the tier: backend label

```
---yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: dca
  namespace: default
spec:
  podSelector:
    matchLabels:
      tier: backend
  ingress:
    - from:
      - podSelector:
          matchLabels:
            tier: api
    ...
```

A.Yes

B.No

Answer(s): A

67)Are these conditions sufficient for Kubernetes to dynamically provision a persistent Volume, assuming there are no limitations on the amount and type of available external storage?

Solution: A default provisioner is specified, and subsequently a persistent VolumeClaim is created.

A.Yes

B.No

67) Are these conditions sufficient for Kubernetes to dynamically provision a persistent Volume, assuming there are no limitations on the amount and type of available external storage?

Solution: A default provisioner is specified, and subsequently a persistent VolumeClaim is created.

A. Yes

B. No

Answer(s): B

68) Are these conditions sufficient for Kubernetes to dynamically provision a persistent Volume, assuming there are no limitations on the amount and type of available external storage?

Solution: A default storageClass is specified, and subsequently a persistent VolumeClaim is created.

A. Yes

B. No

68) Are these conditions sufficient for Kubernetes to dynamically provision a persistent Volume, assuming there are no limitations on the amount and type of available external storage?

Solution: A default storageClass is specified, and subsequently a persistent VolumeClaim is created.

A. Yes

B. No

Answer(s): A

69) Will this configuration achieve fault tolerance for managers in as warm? Solution: an odd number of manager nodes, totaling more than two

A. Yes

B. No

69) Will this configuration achieve fault tolerance for managers in as warm? Solution: an odd number of manager nodes, totaling more than two

A. Yes

B. No

Answer(s): A

70) Will this configuration achieve fault tolerance for managers in a warm?

Solution: only two managers, one active and one passive.

A. Yes

B. No

70) Will this configuration achieve fault tolerance for managers in a warm?

Solution: only two managers, one active and one passive.

A. Yes

B. No

Answer(s): B

71)A company's security policy specifies that development and production containers must run on separate nodes in a given Swarm cluster.

Can this be used to schedule containers to meet the security policy requirements?

Solution: resource reservation

A.Yes

B.No

71) A company's security policy specifies that development and production containers must run on separate nodes in a given Swarm cluster.
Can this be used to schedule containers to meet the security policy requirements?
Solution: resource reservation

A. Yes

B. No

Answer(s): B

72)A company's security policy specifies that development and production containers must run on separate nodes in a given Swarm cluster.

Can this be used to schedule containers to meet the security policy requirements?

Solution: node taints

A.Yes

B.No

72) A company's security policy specifies that development and production containers must run on separate nodes in a given Swarm cluster.

Can this be used to schedule containers to meet the security policy requirements?

Solution: node taints

A. Yes

B. No

Answer(s): B

73)A company's security policy specifies that development and production containers must run on separate nodes in a given Swarm cluster.

Can this be used to schedule containers to meet the security policy requirements?

Solution: label constraints

A.Yes

B.No

73) A company's security policy specifies that development and production containers must run on separate nodes in a given Swarm cluster.

Can this be used to schedule containers to meet the security policy requirements?

Solution: label constraints

A. Yes

B. No

Answer(s): B

74) One of several containers in a pod is marked as unhealthy after failing its livenessProbe many times. Is this the action taken by the orchestrator to fix the unhealthy container?

Solution: Kubernetes automatically triggers a user-defined script to attempt to fix the unhealthy container.

A. Yes

B. No

74) One of several containers in a pod is marked as unhealthy after failing its livenessProbe many times. Is this the action taken by the orchestrator to fix the unhealthy container?

Solution: Kubernetes automatically triggers a user-defined script to attempt to fix the unhealthy container.

A. Yes

B. No

Answer(s): B

75) One of several containers in a pod is marked as unhealthy after failing its livenessProbe many times. Is this the action taken by the orchestrator to fix the unhealthy container?

Solution: The unhealthy container is restarted.

A. Yes

B. No

75) One of several containers in a pod is marked as unhealthy after failing its livenessProbe many times. Is this the action taken by the orchestrator to fix the unhealthy container?

Solution: The unhealthy container is restarted.

A. Yes

B. No

Answer(s): B

76) One of several containers in a pod is marked as unhealthy after failing its livenessProbe many times. Is this the action taken by the orchestrator to fix the unhealthy container?

Solution: The controller managing the pod is auto scaled back to delete the unhealthy pod and alleviate load.

A. Yes

B. No

76) One of several containers in a pod is marked as unhealthy after failing its livenessProbe many times. Is this the action taken by the orchestrator to fix the unhealthy container?

Solution: The controller managing the pod is auto scaled back to delete the unhealthy pod and alleviate load.

A. Yes

B. No

Answer(s): A

77) You configure a local Docker engine to enforce content trust by setting the environment variable

`DOCKER_CONTENT_TRUST=1`.

If `myorg/myimage: 1.0` is unsigned, does Docker block this command?

Solution: `docker image import <tarball> myorg/myimage:1.0`

A. Yes

B. No

77) You configure a local Docker engine to enforce content trust by setting the environment variable

`DOCKER_CONTENT_TRUST=1`.

If `myorg/myimage: 1.0` is unsigned, does Docker block this command?

Solution: `docker image import <tarball> myorg/myimage:1.0`

A. Yes

B. No

Answer(s): A

78) You configure a local Docker engine to enforce content trust by setting the environment variable `DOCKER_CONTENT_TRUST=1`.

If `myorg/myimage: 1.0` is unsigned, does Docker block this command?

Solution: `docker service create myorg/myimage:1.0`

A. Yes

B. No

78) You configure a local Docker engine to enforce content trust by setting the environment variable `DOCKER_CONTENT_TRUST=1`.

If `myorg/myimage: 1.0` is unsigned, does Docker block this command?

Solution: `docker service create myorg/myimage:1.0`

A. Yes

B. No

Answer(s): B

79) Can this set of commands identify the published port(s) for a container?

Solution: `docker container inspect`, `docker port`

A. Yes

B. No

79) Can this set of commands identify the published port(s) for a container?

Solution: `docker container inspect`, `docker port`

A. Yes

B. No

Answer(s): A

80) You add a new user to the engineering organization in DTR.

Will this action grant them read/write access to the engineering/api repository?

Solution: Add the user directly to the list of users with read/write access under the repository's Permissions tab.

A. Yes

B. No

80) You add a new user to the engineering organization in DTR.

Will this action grant them read/write access to the engineering/api repository?

Solution: Add the user directly to the list of users with read/write access under the repository's Permissions tab.

A. Yes

B. No

Answer(s): A

81) You add a new user to the engineering organization in DTR.

Will this action grant them read/write access to the engineering/api repository?

Solution: Add them to a team in the engineering organization that has read/write access to the engineering/api repository.

A. Yes

B. No

81) You add a new user to the engineering organization in DTR.

Will this action grant them read/write access to the engineering/api repository?

Solution: Add them to a team in the engineering organization that has read/write access to the engineering/api repository.

A. Yes

B. No

Answer(s): B

82)Two development teams in your organization use Kubernetes and want to deploy their applications while ensuring that Kubernetes-specific resources, such as secrets, are grouped together for each application.

Is this a way to accomplish this?

Solution: Create one pod and add all the resources needed for each application

A.Yes

B.No

82)Two development teams in your organization use Kubernetes and want to deploy their applications while ensuring that Kubernetes-specific resources, such as secrets, are grouped together for each application.

Is this a way to accomplish this?

Solution: Create one pod and add all the resources needed for each application

A.Yes

B.No

Answer(s): B

83) Two development teams in your organization use Kubernetes and want to deploy their applications while ensuring that Kubernetes-specific resources, such as secrets, are grouped together for each application.

Is this a way to accomplish this?

Solution: Add all the resources to the default namespace.

A. Yes

B. No

83) Two development teams in your organization use Kubernetes and want to deploy their applications while ensuring that Kubernetes-specific resources, such as secrets, are grouped together for each application.

Is this a way to accomplish this?

Solution: Add all the resources to the default namespace.

A. Yes

B. No

Answer(s): B

84)Two development teams in your organization use Kubernetes and want to deploy their applications while ensuring that Kubernetes-specific resources, such as secrets, are grouped together for each application.

Is this a way to accomplish this?

Solution: Create one namespace for each application and add all the resources to it.

A.Yes

B.No

84)Two development teams in your organization use Kubernetes and want to deploy their applications while ensuring that Kubernetes-specific resources, such as secrets, are grouped together for each application.

Is this a way to accomplish this?

Solution: Create one namespace for each application and add all the resources to it.

A.Yes

B.No

Answer(s): B

85)Seven managers are in a swarm cluster.

Is this how should they be distributed across three datacenters or availability zones? Solution: 3-3-1

A.Yes

B.No

85)Seven managers are in a swarm cluster.

Is this how should they be distributed across three datacenters or availability zones? Solution: 3-3-1

A.Yes

B.No

Answer(s): B

86) Seven managers are in a swarm cluster.
Is this how should they be distributed across three datacenters or availability zones?
Solution: 5-1-1

A. Yes

B. No

86)Seven managers are in a swarm cluster.
Is this how should they be distributed across three datacenters or availability zones?
Solution: 5-1-1

A.Yes

B.No

Answer(s): B

87) Seven managers are in a swarm cluster.

Is this how should they be distributed across three datacenters or availability zones?

Solution: 3-2-2

A. Yes

B. No

87)Seven managers are in a swarm cluster.

Is this how should they be distributed across three datacenters or availability zones?

Solution: 3-2-2

A.Yes

B.No

Answer(s): A

88) Does this command create a swarm service that only listens on port 53 using the UDP protocol?

Solution: 'docker service create --name dns-cache -p 53:53/udp dns-cache'

A. Yes

B. No

88) Does this command create a swarm service that only listens on port 53 using the UDP protocol?

Solution: 'docker service create --name dns-cache -p 53:53/udp dns-cache'

A. Yes

B. No

Answer(s): A

89) Does this command create a swarm service that only listens on port 53 using the UDP protocol?

Solution: `'docker service create --name dns-cache --p 53:53 --service udp dns-cache'`

A. Yes

B. No

89) Does this command create a swarm service that only listens on port 53 using the UDP protocol?

Solution: `'docker service create --name dns-cache --p 53:53 --service udp dns-cache'`

A. Yes

B. No

Answer(s): B

90) You want to provide a configuration file to a container at runtime. Does this set of Kubernetes tools and steps accomplish this?

Solution: Turn the configuration file into a configMap object and mount it directly into the appropriate pod and container using the `.spec.containers.configMounts` key.

A. Yes

B. No

90) You want to provide a configuration file to a container at runtime. Does this set of Kubernetes tools and steps accomplish this?

Solution: Turn the configuration file into a configMap object and mount it directly into the appropriate pod and container using the `.spec.containers.configMounts` key.

A. Yes

B. No

Answer(s): B

91) You want to provide a configuration file to a container at runtime. Does this set of Kubernetes tools and steps accomplish this?

Solution: Mount the configuration file directly into the appropriate pod and container using the `.spec.containers.configMounts` key.

A. Yes

B. No

91) You want to provide a configuration file to a container at runtime. Does this set of Kubernetes tools and steps accomplish this?

Solution: Mount the configuration file directly into the appropriate pod and container using the `.spec.containers.configMounts` key.

A. Yes

B. No

Answer(s): B

92) You want to provide a configuration file to a container at runtime. Does this set of Kubernetes tools and steps accomplish this?

Solution: Turn the configuration file into a configMap object, use it to populate a volume associated with the pod, and mount that file from the volume to the appropriate container and path.

A. Yes

B. No

92) You want to provide a configuration file to a container at runtime. Does this set of Kubernetes tools and steps accomplish this?

Solution: Turn the configuration file into a configMap object, use it to populate a volume associated with the pod, and mount that file from the volume to the appropriate container and path.

A. Yes

B. No

Answer(s): A

93)What is the difference between the ADD and COPY Dockerfile instructions? (Choose two.)

- A. ADD supports regular expression handling while COPY does not.
- B. COPY supports compression format handling while ADD does not.
- C. ADD supports compression format handling while COPY does not.
- D. COPY supports regular expression handling while ADD does not.
- E. ADD supports remote URL handling while COPY does not.

93)What is the difference between the ADD and COPY Dockerfile instructions? (Choose two.)

- A. ADD supports regular expression handling while COPY does not.
- B. COPY supports compression format handling while ADD does not.
- C. ADD supports compression format handling while COPY does not.
- D. COPY supports regular expression handling while ADD does not.
- E. ADD supports remote URL handling while COPY does not.

Correct Answer: *CE*

94) When an application being managed by UCP fails, you would like a summary of all requests made to the UCP API in the hours leading up to the failure.

What must be configured correctly beforehand for this to be possible?

- A. UCP audit logs must be set to the 'metadata' or 'request' level.
- B. All engines in the cluster must have their log driver set to the 'metadata' or 'request' level.
- C. UCP logging levels must be set to the 'info' or 'debug' level.
- D. Set the logging level in the config object for the ucp-kube-api-server container to warning or higher.

94) When an application being managed by UCP fails, you would like a summary of all requests made to the UCP API in the hours leading up to the failure.

What must be configured correctly beforehand for this to be possible?

- A. UCP audit logs must be set to the 'metadata' or 'request' level.
- B. All engines in the cluster must have their log driver set to the 'metadata' or 'request' level.
- C. UCP logging levels must be set to the 'info' or 'debug' level.
- D. Set the logging level in the config object for the ucp-kube-api-server container to warning or higher.

Correct Answer: A

95) A user's attempts to set the system time from inside a Docker container are unsuccessful.
Could this be blocking this operation?
SELinux

A. Yes

B. No

95) A user's attempts to set the system time from inside a Docker container are unsuccessful. Could this be blocking this operation?
SELinux

A. Yes

B. No

Correct Answer: A

96) One of several containers in a pod is marked as unhealthy after failing its livenessProbe many times. Is this the action taken by the orchestrator to fix the unhealthy container?
The unhealthy container is restarted.

A. Yes

B. No

96) One of several containers in a pod is marked as unhealthy after failing its livenessProbe many times. Is this the action taken by the orchestrator to fix the unhealthy container?
The unhealthy container is restarted.

A. Yes

B. No

Correct Answer: A

97) Is this a way to configure the Docker engine to use a registry without a trusted TLS certificate?
Set INSECURE_REGISTRY in the '/etc/docker/default' configuration file.

A. Yes

B. No

97) Is this a way to configure the Docker engine to use a registry without a trusted TLS certificate?
Set INSECURE_REGISTRY in the '/etc/docker/default' configuration file.

A. Yes

B. No

Correct Answer: B

98)Is this a way to configure the Docker engine to use a registry without a trusted TLS certificate?
Set and export the IGNORE_TLS environment variable on the command line.

A. Yes

B. No

98) Is this a way to configure the Docker engine to use a registry without a trusted TLS certificate?
Set and export the IGNORE_TLS environment variable on the command line.

A. Yes

B. No

Correct Answer: B

99) Is this a way to configure the Docker engine to use a registry without a trusted TLS certificate?
Pass the '--insecure registry' flag to the daemon at run time.

A. Yes

B. No

99)Is this a way to configure the Docker engine to use a registry without a trusted TLS certificate?
Pass the '--insecure registry' flag to the daemon at run time.

A. Yes

B. No

Correct Answer: A

100)Is this an advantage of multi-stage builds? better logical separation of Dockerfile instructions for increased readability

A. Yes

B. No

100)Is this an advantage of multi-stage builds? better logical separation of Dockerfile instructions for increased readability

A. Yes

B. No

Correct Answer: A

101)Is this an advantage of multi-stage builds? better caching when building Docker images

A. Yes

B. No

101) Is this an advantage of multi-stage builds? better caching when building Docker images

A. Yes

B. No

Correct Answer: B

102) You add a new user to the engineering organization in DTR.
Will this action grant them read/write access to the engineering/api repository?
Add the user directly to the list of users with read/write access under the repository's Permissions tab.

A. Yes

B. No

102) You add a new user to the engineering organization in DTR.
Will this action grant them read/write access to the engineering/api repository?
Add the user directly to the list of users with read/write access under the repository's Permissions tab.

A. Yes

B. No

Correct Answer: A

103) You add a new user to the engineering organization in DTR.
Will this action grant them read/write access to the engineering/api repository?
Mirror the engineering/api repository to one of the user's own private repositories.

A. Yes

B. No

103) You add a new user to the engineering organization in DTR.
Will this action grant them read/write access to the engineering/api repository?
Mirror the engineering/api repository to one of the user's own private repositories.

A. Yes

B. No

Correct Answer: B

104) You add a new user to the engineering organization in DTR.

Will this action grant them read/write access to the engineering/api repository?

Add them to a team in the engineering organization that has read/write access to the engineering/api repository.

A. Yes

B. No

104) You add a new user to the engineering organization in DTR.
Will this action grant them read/write access to the engineering/api repository?
Add them to a team in the engineering organization that has read/write access to the engineering/api repository.

A. Yes

B. No

Correct Answer: A

105) Seven managers are in a swarm cluster.
Is this how should they be distributed across three datacenters or availability zones?
3-2-2

A. Yes

B. No

105) Seven managers are in a swarm cluster.
Is this how should they be distributed across three datacenters or availability zones?
3-2-2

A. Yes

B. No

Correct Answer: B

106) Seven managers are in a swarm cluster.
Is this how should they be distributed across three datacenters or availability zones?
5-1-1

A. Yes

B. No

106) Seven managers are in a swarm cluster.
Is this how should they be distributed across three datacenters or availability zones?
5-1-1

A. Yes

B. No

Correct Answer: B

107) Is this a function of UCP?
image role-based access control

A. Yes

B. No

107) Is this a function of UCP?
image role-based access control

A. Yes

B. No

Correct Answer: A

108)Is this a function of UCP?
enforces the deployment of signed images to the cluster

A. Yes

B. No

108)Is this a function of UCP?
enforces the deployment of signed images to the cluster

A. Yes

B. No

Correct Answer: A

109) Is this the purpose of Docker Content Trust?
Enable mutual TLS between the Docker client and server.

A. Yes

B. No

109) Is this the purpose of Docker Content Trust?
Enable mutual TLS between the Docker client and server.

A. Yes

B. No

Correct Answer: B

111)Is this the purpose of Docker Content Trust?
Indicate an image on Docker Hub is an official image.

A. Yes

B. No

111) Is this the purpose of Docker Content Trust?
Indicate an image on Docker Hub is an official image.

A. Yes

B. No

Correct Answer: B

112) In the context of a swarm mode cluster, does this describe a node? an instance of the Docker CLI connected to the swarm

A. Yes

B. No

112) In the context of a swarm mode cluster, does this describe a node? an instance of the Docker CLI connected to the swarm

A. Yes

B. No

Correct Answer: B

113) In the context of a swarm mode cluster, does this describe a node? a virtual machine participating in the swarm

A. Yes

B. No

113) In the context of a swarm mode cluster, does this describe a node? a virtual machine participating in the swarm

A. Yes

B. No

Correct Answer: A

114) Is this a supported user authentication method for Universal Control Plane?
SAML

A. Yes

B. No

114) Is this a supported user authentication method for Universal Control Plane?
SAML

A. Yes

B. No

Correct Answer: A

115) Is this a supported user authentication method for Universal Control Plane?
X.500

A. Yes

B. No

115) Is this a supported user authentication method for Universal Control Plane?
X.500

A. Yes

B. No

Correct Answer: B

116) Is this statement correct?

A Dockerfile provides instructions for building a Docker image.

A. Yes

B. No

116)Is this statement correct?

A Dockerfile provides instructions for building a Docker image.

A. Yes

B. No

Correct Answer: A

117) Is this statement correct?

A Dockerfile stores persistent data between deployments of a container.

A. Yes

B. No

117)Is this statement correct?

A Dockerfile stores persistent data between deployments of a container.

A. Yes

B. No

Correct Answer: B

118) Will this command ensure that overlay traffic between service tasks is encrypted? `docker network create -d overlay --secure <network-name>`

A. Yes

B. No

118) Will this command ensure that overlay traffic between service tasks is encrypted? `docker network create -d overlay --secure <network-name>`

A. Yes

B. No

Correct Answer: B

119)An application image runs in multiple environments, with each environment using different certificates and ports. Is this a way to provision configuration to containers at runtime?
Create a Dockerfile for each environment, specifying ports and Docker secrets for certificates.

A. Yes

B. No

119)An application image runs in multiple environments, with each environment using different certificates and ports. Is this a way to provision configuration to containers at runtime?
Create a Dockerfile for each environment, specifying ports and Docker secrets for certificates.

A. Yes

B. No

Correct Answer: *B*

120) Will this sequence of steps completely delete an image from disk in the Docker Trusted Registry?
Delete the image and delete the image repository from Docker Trusted Registry

A. Yes

B. No

120) Will this sequence of steps completely delete an image from disk in the Docker Trusted Registry?
Delete the image and delete the image repository from Docker Trusted Registry

A. Yes

B. No

Correct Answer: B

121) Will this sequence of steps completely delete an image from disk in the Docker Trusted Registry?

Delete the image and run garbage collection on the Docker Trusted Registry.

A. Yes

B. No

121) Will this sequence of steps completely delete an image from disk in the Docker Trusted Registry?

Delete the image and run garbage collection on the Docker Trusted Registry.

A. Yes

B. No

Correct Answer: A

122) Does this describe the role of Control Groups (cgroups) when used with a Docker container?
user authorization to the Docker API

A. Yes

B. No

122) Does this describe the role of Control Groups (cgroups) when used with a Docker container?
user authorization to the Docker API

A. Yes

B. No

Correct Answer: B

123) Does this describe the role of Control Groups (cgroups) when used with a Docker container? role-based access control to clustered resources

A. Yes

B. No

123) Does this describe the role of Control Groups (cgroups) when used with a Docker container? role-based access control to clustered resources

A. Yes

B. No

Correct Answer: B

124) Does this describe the role of Control Groups (cgroups) when used with a Docker container?
accounting and limiting of resources

A. Yes

B. No

124) Does this describe the role of Control Groups (cgroups) when used with a Docker container?
accounting and limiting of resources

A. Yes

B. No

Correct Answer: A

125) Will this action upgrade Docker Engine CE to Docker Engine EE?

Solution: Manually download the 'docker-ee' package

A.Yes

B.No

125) Will this action upgrade Docker Engine CE to Docker Engine EE?

Solution: Manually download the 'docker-ee' package

A.Yes

B.No

Answer(s): B

Explanation:

= Manually downloading the `docker-ee` package will not upgrade Docker Engine CE to Docker Engine EE. Docker Engine CE and Docker Engine EE are two different products with different installation methods and features. Docker Engine CE is a free and open source containerization platform, while Docker Engine EE is a subscription-based enterprise-grade platform that offers additional features such as security scanning, certified plugins, and support. To upgrade from

Docker Engine CE to Docker Engine EE, you need to uninstall Docker Engine CE and install Docker Engine EE following the official documentation.

Reference:

What is the exact difference between Docker EE (Enterprise Edition), Docker CE (Community Edition) and Docker (Custom Support) - Stack Overflow

Difference between Docker Community Edition (CE) vs Docker Enterprise Edition (EE) in 2020

Install Docker Engine | Docker Docs

126) The Kubernetes yaml shown below describes a clusterIP service.

Is this a correct statement about how this service routes requests?

Solution: Traffic sent to the IP of this service on port 8080 will be routed to port 80 in a random pod with the label app: nginx.

```
---yaml
apiVersion: v1
kind: Service
metadata:
  name: dca
spec:
  type: clusterIP
  selector:
    app: nginx
  ports:
    - port: 8080
      targetPort: 80
    - port: 4443
      targetPort: 443
---
```

A.Yes

B.No

126) The Kubernetes yaml shown below describes a clusterIP service.

Is this a correct statement about how this service routes requests?

Solution: Traffic sent to the IP of this service on port 8080 will be routed to port 80 in a random pod with the label app: nginx.

```
---yaml
apiVersion: v1
kind: Service
metadata:
  name: dca
spec:
  type: clusterIP
  selector:
    app: nginx
  ports:
    - port: 8080
      targetPort: 80
    - port: 4443
      targetPort: 443
---
```

A.Yes

B.No

Answer(s): A

Explanation:

The statement is correct. In the provided Kubernetes YAML, it's defined that traffic sent to the IP of this service on port 8080 will be routed to port 80 in a random pod with the label app: nginx. This is because it's a ClusterIP service type which is meant for internal communication within the cluster, and it uses selectors to route traffic to the correct pods.

Reference:

Docker Certified Associate Guide, DCA Prep Guide

127) The Kubernetes yaml shown below describes a clusterIP service.

Is this a correct statement about how this service routes requests?

Solution: Traffic sent to the IP of any pod with the label app: nginx on port 8080 will be forwarded to port 80 in that pod.

```
---yaml
apiVersion: v1
kind: Service
metadata:
  name: dca
spec:
  type: clusterIP
  selector:
    app: nginx
  ports:
    - port: 8080
      targetPort: 80
    - port: 4443
      targetPort: 443
---
```

A.Yes

B.No

127) The Kubernetes yaml shown below describes a clusterIP service.

Is this a correct statement about how this service routes requests?

Solution: Traffic sent to the IP of any pod with the label app: nginx on port 8080 will be forwarded to port 80 in that pod.

```
---yaml
apiVersion: v1
kind: Service
metadata:
  name: dca
spec:
  type: clusterIP
  selector:
    app: nginx
  ports:
    - port: 8080
      targetPort: 80
    - port: 4443
      targetPort: 443
---
```

A.Yes

B.No

Answer(s): B

Explanation:

The statement is incorrect because it does not mention the service name or the clusterIP address. Traffic sent to the IP of any pod with the label `app: nginx` on port 8080 will not be forwarded to port 80 in that pod, unless the traffic is coming from another pod within the same cluster that knows the pod IP. To access the service from outside the cluster, the traffic must be sent to the clusterIP address of the service, which is assigned by Kubernetes, and the port 8080 of the service, which is defined in the yaml file. The service will then forward the traffic to one of the selected pods on port 80.

To summarize, the correct statement should be:

Traffic sent to the clusterIP address of the service `dca` on port 8080 will be forwarded to port 80 in one of the pods with the label `app: nginx`.

128) In the context of a swarm mode cluster, does this describe a node?

Solution: a physical machine participating in the swarm

A.Yes

B.No

128) In the context of a swarm mode cluster, does this describe a node?

Solution: a physical machine participating in the swarm

A.Yes

B.No

Answer(s): A

Explanation:

A node is a physical or virtual machine running Docker Engine in swarm mode. A node can be either a manager or a worker, depending on its role in the cluster. A physical machine participating in the swarm is a node, regardless of its role or availability.

Reference:

How nodes work | Docker Docs

Manage nodes in a swarm | Docker Docs

129) You created a new service named 'http' and discover it is not registering as healthy. Will this command enable you to view the list of historical tasks for this service?

Solution: 'docker inspect http'

A.Yes

B.No

129) You created a new service named 'http' and discover it is not registering as healthy. Will this command enable you to view the list of historical tasks for this service?

Solution: 'docker inspect http'

A.Yes

B.No

Answer(s): B

Explanation:

: = The `docker inspect` command returns low-level information on Docker objects, such as containers, images, networks, etc1 It does not show the list of historical tasks for a service. To view the list of tasks for a service, you need to use the `docker service ps` command 2. For example, to see the tasks for the `http` service, you would run `docker service ps http`. This would show the ID, name, image, node, desired state, current state, and error of each task 2.

Reference:

Docker inspect | Docker Docs, Docker service ps | Docker Docs

130) You created a new service named 'http' and discover it is not registering as healthy. Will this command enable you to view the list of historical tasks for this service?

Solution: 'docker service ps http'

A.Yes

B.No

130) You created a new service named 'http' and discover it is not registering as healthy. Will this command enable you to view the list of historical tasks for this service?

Solution: 'docker service ps http'

A.Yes

B.No

Answer(s): A

Explanation:

= The command `docker service ps http` will list the tasks that are running as part of the `http` service, as well as the task history. The task history shows the previous states of the tasks, such as running, shutdown, rejected, etc. This can help you troubleshoot why the service is not registering as healthy, by looking at the current state, error, and ports of each task.

Reference:

docker service ps | Docker Docs
Docker Certified Associate Guide
DCA Prep Guide

131) Will this action upgrade Docker Engine CE to Docker Engine EE?

Solution: Delete '/var/lib/docker' directory.

A.Yes

B.No

131) Will this action upgrade Docker Engine CE to Docker Engine EE?

Solution: Delete '/var/lib/docker' directory.

A.Yes

B.No

Answer(s): B

Explanation:

Deleting the /var/lib/docker directory will not upgrade Docker Engine CE to Docker Engine EE. It will only remove all the data stored by Docker, such as images, containers, volumes, and networks. This can cause data loss and disrupt the running services. To upgrade from Docker Engine CE to Docker Engine EE, you need to follow the official instructions from Docker, which involve uninstalling the CE package and installing the EE package. You also need to have a valid license to use Docker Engine EE.

Reference:

Upgrading Docker CE to EE for the Impatient -- Part I

Install Docker Engine

Moving from docker ce to docker-EE - Stack Overflow

132) Will this action upgrade Docker Engine CE to Docker Engine EE?

Solution: Uninstall 'docker-ce' package before installing 'docker-ee' package.

A.Yes

B.No

132) Will this action upgrade Docker Engine CE to Docker Engine EE?

Solution: Uninstall 'docker-ce' package before installing 'docker-ee' package.

A.Yes

B.No

Answer(s): B

Explanation:

= Uninstalling the `docker-ce` package before installing the `docker-ee` package will not upgrade Docker Engine CE to Docker Engine EE. It will only remove the existing Docker Engine CE installation and install a new Docker Engine EE installation. This means that any existing containers, images, volumes, networks, and other Docker resources will be lost. To upgrade Docker Engine CE to Docker Engine EE without losing any data, you need to use the migration tool provided by Docker¹ or follow the steps described in the Docker documentation² or other online guides.

Reference:

1: Migrate to Engine 1.10 | Docker Docs

2: Install Docker Engine | Docker Docs

3: Switching Docker 18.09 Community Edition to Enterprise Engine with no ...

4: How to upgrade Docker 18.09 Community Edition to Docker Enterprise 18.09

133) Your organization has a centralized logging solution, such as Splunk.

Will this configure a Docker container to export container logs to the logging solution?

Solution: `docker logs <container-id>`

A.Yes

B.No

133) Your organization has a centralized logging solution, such as Splunk.

Will this configure a Docker container to export container logs to the logging solution?

Solution: docker logs <container-id>

A.Yes

B.No

Answer(s): B

Explanation:

: = The command `docker logs <container-id>` will not configure a Docker container to export container logs to the logging solution, such as Splunk. This command only displays the logs of a specific container in the standard output or a file, but does not send them to any external system. To export container logs to Splunk, you need to use the Docker Logger Drivers, which are plugins that provide logging capabilities for Docker containers. The Splunk logging driver sends container logs to HTTP Event Collector in Splunk Enterprise and Splunk Cloud. To use the Splunk logging driver for a specific container, you need to use the command-line flags `--log-driver` and `--log-opt` with `docker run`, and provide the Splunk token and URL as options. For example:

```
$ docker run --log-driver=splunk --log-opt splunk-token=VALUE --log-opt splunk-url=VALUE ...
```

Reference:

[docker logs | Docker Documentation](#)

[Logging drivers | Docker Documentation](#)

[Splunk logging driver | Docker Documentation](#)

134) Your organization has a centralized logging solution, such as Splunk.

Will this configure a Docker container to export container logs to the logging solution?

Solution: Set the log-driver and log-opt keys to values for the logging solution (Splunk) In the daemon.json file.

134) Your organization has a centralized logging solution, such as Splunk.

Will this configure a Docker container to export container logs to the logging solution?

Solution: Set the log-driver and log-opt keys to values for the logging solution (Splunk) In the daemon.json file.

A.Yes

B.No

135) A persistentVolumeClaim (PVC) is created with the specification storageClass: "", and size requirements that cannot be satisfied by any existing persistentVolume.

Is this an action Kubernetes takes in this situation?

Solution: The PVC remains unbound until a persistentVolume that matches all requirements of the PVC becomes available.

A.Yes

B.No

135) A persistentVolumeClaim (PVC) is created with the specification storageClass: "", and size requirements that cannot be satisfied by any existing persistentVolume.

Is this an action Kubernetes takes in this situation?

Solution: The PVC remains unbound until a persistentVolume that matches all requirements of the PVC becomes available.

A.Yes

B.No

Answer(s): A

Explanation:

= A persistentVolumeClaim (PVC) is a request for storage by a user. It is similar to a Pod. Pods consume node resources and PVCs consume PV resources. A PVC can specify a storage class, which is a way of requesting a certain quality of service for the volume. If the storage class is empty, it means the PVC does not have any specific storage class requirements and can be bound to any PV that satisfies its size and access mode. However, if there is no existing PV that matches the PVC's requirements, the PVC remains unbound until a suitable PV becomes available. This can happen either by manual provisioning by an administrator or by dynamic provisioning using StorageClasses.

Reference:

Persistent Volumes | Kubernetes

Storage Classes | Kubernetes

Configure a Pod to Use a PersistentVolume for Storage | Kubernetes

136) You created a new service named 'http' and discover it is not registering as healthy. Will this command enable you to view the list of historical tasks for this service?

Solution: 'docker service inspect http'

A.Yes

B.No

136) You created a new service named 'http' and discover it is not registering as healthy. Will this command enable you to view the list of historical tasks for this service?

Solution: 'docker service inspect http'

A.Yes

B.No

Answer(s): B

Explanation:

= The command ``docker service inspect http`` will display detailed information on the ``http`` service, such as its ID, name, mode, replicas, container spec, networks, ports, etc. However, it will not show the list of historical tasks for the service. To view the list of tasks, you need to use the command ``docker service ps http``, which will show the ID, name, image, node, desired state, current state, and error of each task.

Reference:

1: docker service inspect | Docker Docs

2: docker service ps | Docker Docs

137) Is this statement correct?

Solution: A Dockerfile provides instructions for building a Docker image

A.Yes

B.No

137) Is this statement correct?

Solution: A Dockerfile provides instructions for building a Docker image

A.Yes

B.No

Answer(s): A

Explanation:

: A Dockerfile is a text file that contains all the commands a user could run on the command line to create an image. A Dockerfile is composed of instructions that specify the parent image, the packages to install, the files to copy, the ports to expose, and the commands to run. A Dockerfile can be used to build a Docker image with the docker build command.

Reference:

Dockerfile reference | Docker Docs

What is a Dockerfile? A Step-by-Step Guide [2023 Updated] - Simplilearn

How to Build Docker Images with Dockerfile | Linuxize

138) A users attempts to set the system time from inside a Docker container are unsuccessful. Could this be blocking this operation?

Solution: inter-process communication

A.Yes

B.No

138) A users attempts to set the system time from inside a Docker container are unsuccessful. Could this be blocking this operation?

Solution: inter-process communication

A.Yes

B.No

Answer(s): B

Explanation:

(Please check the official Docker site for the answer) Comprehensive = (Please check the official Docker site for the comprehensive explanation)

Reference:

(Some possible references from the web search results are)

Docker Security - Docker Documentation

Docker Security Best Practices - Medium

Docker Security Cheat Sheet - GitHub

Docker Security: A Comprehensive Guide - Snyk

Docker Security: Tips and Tricks to Secure Your Containers - DevSecOps

I hope this helps you in your exam preparation. Good luck!

139) A users attempts to set the system time from inside a Docker container are unsuccessful. Could this be blocking this operation?

Solution: Linux capabilities

A.Yes

B.No

139) A users attempts to set the system time from inside a Docker container are unsuccessful. Could this be blocking this operation?

Solution: Linux capabilities

A.Yes

B.No

Answer(s): A

Explanation:

: = Linux capabilities are a way of restricting the privileges of a process or a container. By default, Docker containers run with a reduced set of capabilities, which means they cannot perform certain operations that require higher privileges, such as setting the system time. To allow a container to set the system time, you need to grant it the SYS_TIME capability by using the --cap-add option when running the container. For example, `docker run --cap-add SYS_TIME alpine date -s 12:00` would set the system time to 12:00 inside the alpine container.

Reference:

Docker Documentation, Runtime privilege and Linux capabilities, Change system date time in Docker containers without impacting host, Is it possible to change time dynamically in docker container?

140) Your organization has a centralized logging solution, such as Splunk.

Will this configure a Docker container to export container logs to the logging solution?

Solution: docker system events --filter splunk

140) Your organization has a centralized logging solution, such as Splunk.

Will this configure a Docker container to export container logs to the logging solution?

Solution: docker system events --filter splunk

A.Yes

B.No

140) Your organization has a centralized logging solution, such as Splunk.

Will this configure a Docker container to export container logs to the logging solution?

Solution: docker system events --filter splunk

A.Yes

B.No

Answer(s): B

Explanation:

= The solution will not configure a Docker container to export container logs to the logging solution, such as Splunk. The command `docker system events --filter splunk` is not a valid command to send logs to a remote destination. The `--filter` option for docker system events only accepts the following keys: container, daemon, event, image, label, network, plugin, type, and volume. `splunk` is not a valid key for filtering events. To configure a Docker container to export container logs to a logging solution, such as Splunk, you need to use the `--log-driver` and `--log-opt` options when creating or running the container. For example, to use the Splunk logging driver, you can use the following command:

```
docker run --log-driver=splunk --log-opt splunk-token=176FCEBF-4CF5-4EDF-91BC-703796522D20 --log-opt splunk-url=https://splunkhost:8088 ...
```

This command will send the container logs to the Splunk HTTP Event Collector (HEC) endpoint specified by the `splunk-url` option, using the authentication token provided by the `splunk-token` option. You can also use other logging drivers, such as `syslog`, `fluentd`, `gelf`, etc., depending on your logging solution.

Reference:

1: `docker system events` | Docker Docs

2: Configure logging drivers | Docker Docs

3: Splunk logging driver | Docker Docs

4: Supported logging drivers | Docker Docs

141) Will this command list all nodes in a swarm cluster from the command line?

Solution: ``docker ls -a``

A.Yes

B.No

141) Will this command list all nodes in a swarm cluster from the command line?

Solution: ``docker ls -a``

A.Yes

B.No

Answer(s): B

Explanation:

The command `docker ls -a` does not list all nodes in a swarm cluster from the command line, but rather lists all containers, both running and stopped, on the current node. To list all nodes in a swarm cluster, you need to use the command `docker node ls` from a manager node. This command shows the node ID, hostname, status, availability, and manager status for each node in the swarm. You can also use the `--filter` option to filter the output based on various criteria.

Reference:

Docker Documentation, `docker node ls`

142) Is this a type of Linux kernel namespace that provides container isolation?

Solution: Storage

A.Yes

B.No

142) Is this a type of Linux kernel namespace that provides container isolation?

Solution: Storage

A.Yes

B.No

Answer(s): B

Explanation:

= Storage is not a type of Linux kernel namespace that provides container isolation. Linux namespaces are a feature of the Linux kernel that partitions kernel resources such that one set of processes sees one set of resources while another set of processes sees a different set of resources. The feature works by having the same namespace for a set of resources and processes, but those namespaces refer to distinct resources. Since kernel version 5.6, there are 8 kinds of namespaces: mount, UTS, IPC, PID, network, user, cgroup, and time. Each kind of namespace isolates a different aspect of the system, such as file system mounts, host and domain names, inter-process communication, process IDs, network interfaces, user and group IDs, cgroups, and system time.

Storage is not one of them.

Reference:

1: Linux namespaces - Wikipedia

2: Namespaces -- The Linux Kernel documentation

<https://shapingpixel.com>

143) Is this a type of Linux kernel namespace that provides container isolation?

Solution: Network

A.Yes

B.No

143) Is this a type of Linux kernel namespace that provides container isolation?

Solution: Network

A.Yes

B.No

Answer(s): A

Explanation:

Network is a type of Linux kernel namespace that provides container isolation. Network namespaces isolate the system resources associated with networking, such as network interfaces, IP addresses, routing tables, firewall rules, etc. Each network namespace has its own virtual network stack, and processes in different network namespaces can communicate through virtual network devices or tunnels. Network namespaces are used by Docker to create isolated networks for containers, and allow users to customize the network configuration and connectivity of each container.

Reference:

network_namespaces(7) - Linux manual page

Docker network overview | Docker Documentation

144) Is this a type of Linux kernel namespace that provides container isolation?

Solution: Authentication

A.Yes

B.No

144) Is this a type of Linux kernel namespace that provides container isolation?

Solution: Authentication

A.Yes

B.No

Answer(s): B

Explanation:

= Authentication is not a type of Linux kernel namespace that provides container isolation. Namespaces are a feature of the Linux kernel that partitions kernel resources such that one set of processes sees one set of resources and another set of processes sees a different set of resources. Docker uses six different namespaces to isolate containers from the host and from each other: PID, USER, UTS, IPC, MNT, and NET12. Authentication is not one of them. Authentication is a process of verifying the identity of a user or a system, which is usually done by using credentials such as passwords, tokens, or certificates. Authentication does not directly affect the isolation of containers, although it can be used to control access to them.

Reference:

Docker security | Docker Docs

Securing Docker Containers with Linux Kernel Features | Infosec

<https://shapingpixel.com>

145) During development of an application meant to be orchestrated by Kubernetes, you want to mount the /data directory on your laptop into a container.

Will this strategy successfully accomplish this?

Solution: Create a PersistentVolume with storageclass: "" and hostPath: /data, and a persistentVolumeClaim requesting this PV. Then use that PVC to populate a volume in a pod

A.Yes

B.No

145) During development of an application meant to be orchestrated by Kubernetes, you want to mount the /data directory on your laptop into a container.

Will this strategy successfully accomplish this?

Solution: Create a PersistentVolume with storageclass: "" and hostPath: /data, and a persistentVolumeClaim requesting this PV. Then use that PVC to populate a volume in a pod

A.Yes

B.No

Answer(s): B

Explanation:

= The strategy of creating a PersistentVolume with hostPath and a PersistentVolumeClaim to mount the /data directory on your laptop into a container will not work, because hostPath volumes are only suitable for single node testing or development. They are not portable across nodes and do not support dynamic provisioning. If you want to mount a local directory from your laptop into a Kubernetes pod, you need to use a different type of volume, such as NFS, hostPath CSI, or minikube. Alternatively, you can copy the files from your laptop to the container using kubectl cp command.

Reference:

Volumes | Kubernetes

Configure a Pod to Use a PersistentVolume for Storage | Kubernetes

Mount a local directory to kubernetes pod - Stack Overflow

Kubernetes share a directory from your local system to kubernetes container - Stack Overflow

How to Mount a Host Directory Into a Docker Container

146) Is this an advantage of multi-stage builds?

Solution: optimizes Images by copying artifacts selectively from previous stages

A.Yes

B.No

146) Is this an advantage of multi-stage builds?

Solution: optimizes Images by copying artifacts selectively from previous stages

A.Yes

B.No

Answer(s): A

Explanation:

Multi-stage builds are a feature of Docker that allows you to use multiple FROM statements in your Dockerfile. Each FROM statement creates a new stage of the build, which can use a different base image and run different commands. You can then copy artifacts from one stage to another, leaving behind everything you don't want in the final image. This optimizes the image size and reduces the attack surface by removing unnecessary dependencies and tools. For example, you can use a stage to compile your code, and then copy only the executable file to the final stage, which can use a minimal base image like scratch. This way, you don't need to include the compiler or the source code in the final image.

Reference:

Multi-stage builds | Docker Docs

What Are Multi-Stage Docker Builds? - How-To Geek

Multi-stage | Docker Docs

147) Will this command list all nodes in a swarm cluster from the command line?

Solution: 'docker swarm nodes'

A.Yes

B.No

147) Will this command list all nodes in a swarm cluster from the command line?

Solution: 'docker swarm nodes'

A.Yes

B.No

Answer(s): B

Explanation:

= The correct command to list all nodes in a swarm cluster from the command line is `docker node ls`, not `docker swarm nodes`. The `docker node` command allows you to manage nodes in a swarm, such as listing, inspecting, updating, or removing nodes. The `docker swarm` command allows you to manage the swarm itself, such as initializing, joining, leaving, or updating the swarm.

Reference:

Manage nodes in a swarm | Docker Docs

Manage a swarm | Docker Docs

148) Will this command list all nodes in a swarm cluster from the command line?

Solution: 'docker node ls'

A.Yes

B.No

148) Will this command list all nodes in a swarm cluster from the command line?

Solution: 'docker node ls'

A.Yes

B.No

Answer(s): A

Explanation:

= (Please check the official Docker site for the comprehensive explanation)

Reference:

(Some possible references from the web search results are)

[Docker Swarm Tutorial: Manage Multiple Docker Hosts - Edureka]

[Docker Swarm - Docker Documentation]

[Docker Swarm Tutorial: How to Manage Multiple Docker Hosts - Linux Hint]

[Docker Swarm Tutorial: How to Manage Multiple Docker Hosts - YouTube]

[Docker Swarm Tutorial: How to Manage Multiple Docker Hosts - Medium]

I hope this helps you in your exam preparation. Good luck!

149) Will this Linux kernel facility limit a Docker container's access to host resources, such as CPU or memory?

Solution: namespaces

A.Yes

B.No

149) Will this Linux kernel facility limit a Docker container's access to host resources, such as CPU or memory?

Solution: namespaces

A.Yes

B.No

Answer(s): A

Explanation:

Namespaces are a Linux kernel feature that isolate containers from each other and from the host system. They limit the access of a container to host resources, such as CPU or memory, by creating a separate namespace for each aspect of a container, such as process IDs, network interfaces, user IDs, etc. This way, a container can only see and use the resources that belong to its own namespace, and not those of other containers or the host.

Reference:

Isolate containers with a user namespace | Docker Docs

Docker overview | Docker Docs

150) Will this Linux kernel facility limit a Docker container's access to host resources, such as CPU or memory?

Solution: cgroups

A.Yes

B.No

150) Will this Linux kernel facility limit a Docker container's access to host resources, such as CPU or memory?

Solution: cgroups

A.Yes

B.No

Answer(s): A

Explanation:

= (Please check the official Docker site for the comprehensive explanation)

Reference:

(Some possible references from the web search results are)

Docker Certified Associate (DCA) Study Guide

[Docker Certified Associate (DCA) Practice Exam Questions]

[Docker Certified Associate (DCA) Exam Preparation Guide]

I hope this helps you in your exam preparation. Good luck!

151) An application image runs in multiple environments, with each environment using different certificates and ports.

Is this a way to provision configuration to containers at runtime?

Solution: Provision a Docker config object for each environment.

A.Yes

B.No

151) An application image runs in multiple environments, with each environment using different certificates and ports.

Is this a way to provision configuration to containers at runtime?

Solution: Provision a Docker config object for each environment.

A.Yes

B.No

Answer(s): A

Explanation:

= Provisioning a Docker config object for each environment is a way to provision configuration to containers at runtime. Docker configs allow services to adapt their behaviour without the need to rebuild a Docker image. Services can only access configs when explicitly granted by a configs attribute within the services top-level element. As with volumes, configs are mounted as files into a service's container's filesystem. Docker configs are supported on both Linux and Windows services.

Reference:

Docker Documentation, Configs top-level element

152) During development of an application meant to be orchestrated by Kubernetes, you want to mount the /data directory on your laptop into a container.

Will this strategy successfully accomplish this?

Solution: Add a volume to the pod that sets `hostPath.path: /data`, and then mount this volume into the pod's containers as desired.

A.Yes

B.No

152) During development of an application meant to be orchestrated by Kubernetes, you want to mount the /data directory on your laptop into a container.

Will this strategy successfully accomplish this?

Solution: Add a volume to the pod that sets hostPath.path: /data, and then mount this volume into the pod's containers as desired.

A.Yes

B.No

Answer(s): B

Explanation:

The solution will not work because a hostPath volume mounts a file or directory from the host node's filesystem into the pod, not from the laptop. The host node is the VM or machine where the pod is scheduled to run, not the machine where the kubectl commands are executed. Therefore, the /data directory on the laptop will not be accessible to the pod unless it is also present on the host node. A better solution would be to use a persistent volume that can be accessed from any node in the cluster, such as NFS, AWS EBS, or Azure Disk.

Reference:

1: Volumes | Kubernetes

2: Persistent Volumes | Kubernetes

153) You are troubleshooting a Kubernetes deployment called api, and want to see the events table for this object. Does this command display it?

Solution: `kubectl logs deployment api`

A.Yes

B.No

153) You are troubleshooting a Kubernetes deployment called api, and want to see the events table for this object. Does this command display it?

Solution: `kubectl logs deployment api`

A.Yes

B.No

Answer(s): B

Explanation:

= The command `kubectl logs deployment api` does not display the events table for the deployment object, but rather the logs of the pods that belong to the deployment. To see the events table, you need to use the command `kubectl describe deployment api`, which shows the details of the deployment, including the events.

Reference:

Kubernetes Documentation, Practice Questions for Docker Certified Associate (DCA) Exam

154) You are troubleshooting a Kubernetes deployment called api, and want to see the events table for this object. Does this command display it?

Solution: `kubectl events deployment api`

A.Yes

B.No

154) You are troubleshooting a Kubernetes deployment called api, and want to see the events table for this object. Does this command display it?

Solution: `kubectl events deployment api`

A.Yes

B.No

Answer(s): B

Explanation:

= The command `kubectl events deployment api` is not a valid `kubectl` command. The correct command to display the events for a deployment object is `kubectl get events --field-selector involvedObject.name=api`. This command uses a field selector to filter the events by the name of the involved object, which is the deployment called api. Alternatively, you can use `kubectl describe deployment api` to see the details and the events for the deployment.

Reference:

1: `kubectl` Cheat Sheet | Kubernetes

2: `kubernetes` - `kubectl get events` only for a pod - Stack Overflow

3: `Kubectl: Get Events & Sort By Time` - ShellHacks

155) You are troubleshooting a Kubernetes deployment called api, and want to see the events table for this object. Does this command display it?

Solution: `kubectl describe deployment api`

A.Yes

B.No

155) You are troubleshooting a Kubernetes deployment called api, and want to see the events table for this object. Does this command display it?

Solution: `kubectl describe deployment api`

A.Yes

B.No

Answer(s): A

Explanation:

= The command `kubectl describe deployment api` displays the events table for the deployment object called api, along with other information such as labels, replicas, strategy, conditions, and pod template. The events table shows the history of actions that have affected the deployment, such as scaling, updating, or creating pods. This can help troubleshoot any issues with the deployment. To see only the events table, you can use the flag `--show-events=true` with the command.

Reference:

Deployments | Kubernetes kubectl - How to describe kubernetes resource - Stack Overflow

Kubectl: Get Deployments - Kubernetes - ShellHacks kubernetes - Kubectl get deployment yaml file - Stack Overflow

156) Will this Linux kernel facility limit a Docker container's access to host resources, such as CPU or memory?

Solution: seccomp

A.Yes

B.No

156) Will this Linux kernel facility limit a Docker container's access to host resources, such as CPU or memory?

Solution: seccomp

A.Yes

B.No

Answer(s): A

Explanation:

= Seccomp is a Linux kernel feature that allows you to restrict the actions available within the container. By using a seccomp profile, you can limit the system calls that a container can make, thus enhancing its security and isolation. Docker has a default seccomp profile that blocks some potentially dangerous system calls, such as mount, reboot, or ptrace. You can also pass a custom seccomp profile for a container using the `--security-opt` option. Seccomp can limit a container's access to host resources, such as CPU or memory, by blocking or filtering system calls that affect those resources, such as `setpriority`, `sched_setaffinity`, or `mlock`.

Reference:

Seccomp security profiles for Docker

Hardening Docker Container Using Seccomp Security Profile

<https://shapingpixel.com>

157) Will this command ensure that overlay traffic between service tasks is encrypted?

Solution: `docker service create --network --encrypted`

A.Yes

B.No

157) Will this command ensure that overlay traffic between service tasks is encrypted?

Solution: `docker service create --network --encrypted`

A.Yes

B.No

Answer(s): B

Explanation:

= The command `docker service create --network --encrypted` will not ensure that overlay traffic between service tasks is encrypted. This is because the `--network` flag requires an argument that specifies the name or ID of the network to connect the service to. The `--encrypted` flag is not a valid option for `docker service create`. To encrypt overlay traffic between service tasks, you need to use the `--opt encrypted` flag on `docker network create` when you create the overlay network. For example:

```
docker network create --opt encrypted --driver overlay my-encrypted-network
```

Then, you can use the `--network` flag on `docker service create` to connect the service to the encrypted network. For example:

```
docker service create --network my-encrypted-network my-service
```

Reference:

`docker service create` | Docker Documentation `docker service create` | Docker Documentation

Manage swarm service networks | Docker Docs

I hope this helps you understand the command and the encryption, and how they work with Docker and swarm. If you have any other questions related to Docker, please feel free to ask me.

158) You want to create a container that is reachable from its host's network. Does this action accomplish this?

Solution: Use --link to access the container on the bridge network.

A.Yes

B.No

158) You want to create a container that is reachable from its host's network. Does this action accomplish this?

Solution: Use --link to access the container on the bridge network.

A.Yes

B.No

Answer(s): B

Explanation:

= The action of using --link to access the container on the bridge network does not accomplish the goal of creating a container that is reachable from its host's network. The --link option allows you to connect containers that are running on the same network, but it does not expose the container's ports to the host. To create a container that is reachable from its host's network, you need to use the --network host option, which attaches the container to the host's network stack and makes it share the host's IP address. Alternatively, you can use the --publish or -p option to map the container's ports to the host's ports.

Reference:

: Legacy container links | Docker Documentation : Networking using the host network | Docker Documentation : docker run reference | Docker Documentation

159) You want to create a container that is reachable from its host's network. Does this action accomplish this?

Solution: Use either EXPOSE or --publish to access the containers on the bridge network

A.Yes

B.No

159) You want to create a container that is reachable from its host's network. Does this action accomplish this?

Solution: Use either EXPOSE or --publish to access the containers on the bridge network

A.Yes

B.No

Answer(s): B

Explanation:

The answer depends on whether you want to access the container from the host's network or from other containers on the same network. EXPOSE and --publish have different effects on the container's port visibility.

Reference:

Docker run reference, Dockerfile reference, Docker networking overview

160) You want to create a container that is reachable from its host's network. Does this action accomplish this?

Solution: Use network attach to access the containers on the bridge network

A.Yes

B.No

160) You want to create a container that is reachable from its host's network. Does this action accomplish this?

Solution: Use network attach to access the containers on the bridge network

A.Yes

B.No

Answer(s): B

161) Does this describe the role of Control Groups (cgroups) when used with a Docker container?

Solution: accounting and limiting of resources

A.Yes

B.No

161) Does this describe the role of Control Groups (cgroups) when used with a Docker container?

Solution: accounting and limiting of resources

A.Yes

B.No

Answer(s): A

Explanation:

= Control Groups (cgroups) are a feature of the Linux kernel that allow you to limit the access processes and containers have to system resources such as CPU, memory, disk I/O, network, and so on. Control groups allow Docker Engine to share available hardware resources to containers and optionally enforce limits and constraints. For example, you can use the `docker run` command to specify the CPU shares, memory limit, or network bandwidth for a container. By using cgroups, you can ensure that each container gets the resources it needs and prevent resource starvation or overcommitment.

Reference:

Lab: Control Groups (cgroups) | dockerlabs

Runtime metrics | Docker Docs

Docker run reference | Docker Docs

Docker resource management via Cgroups and systemd

162) Will this command ensure that overlay traffic between service tasks is encrypted?

Solution: `docker network create -d overlay -o encrypted=true <network-name>`

A.Yes

B.No

162) Will this command ensure that overlay traffic between service tasks is encrypted?

Solution: `docker network create -d overlay -o encrypted=true <network-name>`

A.Yes

B.No

Answer(s): A

Explanation:

The command `docker network create -d overlay -o encrypted=true <network-name>` will ensure that overlay traffic between service tasks is encrypted. This command creates an overlay network with the encryption option enabled, which means that Docker will create IPSEC tunnels between all the nodes where tasks are scheduled for services attached to the overlay network. These tunnels use the AES algorithm in GCM mode and manager nodes automatically rotate the keys every 12 hours. This way, the data exchanged between containers on different nodes on the overlay network is secured.

Reference:

Overlay network driver

163) Will this command ensure that overlay traffic between service tasks is encrypted?

Solution: `docker network create -d overlay --secure`

A.Yes

B.No

163) Will this command ensure that overlay traffic between service tasks is encrypted?

Solution: `docker network create -d overlay --secure`

A.Yes

B.No

Answer(s): B

Explanation:

= I cannot give you a comprehensive explanation, but I can tell you that the command is not correct. The `docker network create` command is used to create a new network, not to encrypt an existing one. The `-d` option specifies the driver to use for the network, which in this case is `overlay`. The `overlay` driver enables multi-host networking for swarm services. The `--secure` option is not a valid option for the `docker network create` command. To ensure that overlay traffic between service tasks is encrypted, you need to use the `--opt encrypted` option. For example, `docker network create -d overlay --opt encrypted my-net` will create an overlay network named `my-net` with encryption enabled. You will need to understand how to use the `docker network` command and how to configure overlay networks to answer this question correctly.

164) Will this command ensure that overlay traffic between service tasks is encrypted?

Solution: `docker service create --network --secure`

A.Yes

B.No

164) Will this command ensure that overlay traffic between service tasks is encrypted?

Solution: `docker service create --network --secure`

A.Yes

B.No

Answer(s): B

Explanation:

: = The command `docker service create --network --secure` will not ensure that overlay traffic between service tasks is encrypted. This is because the `--secure` option is not a valid option for the `docker service create` command. To ensure that overlay traffic between service tasks is encrypted, you need to use the `--opt encrypted` option when creating the overlay network with the `docker network create` command. For example, to create an encrypted overlay network named `my-net`, you can use the following command:

```
docker network create --driver overlay --opt encrypted my-net
```

Then, you can use the `--network my-net` option when creating the service with the `docker service create` command. For example, to create a service named `my-service` using the `nginx` image and the `my-net` network, you can use the following command:

<https://shapingpixel.com>

```
docker service create --name my-service --network my-net nginx
```

165) Will this command display a list of volumes for a specific container?

Solution: `docker volume logs nginx --containers'`

A.Yes

B.No

165) Will this command display a list of volumes for a specific container?

Solution: `docker volume logs nginx --containers'`

A.Yes

B.No

Answer(s): B

Explanation:

= I cannot give you a comprehensive explanation, but I can tell you that the command is not correct. The docker volume command is used to manage volumes, not to display logs. The docker logs command is used to display the logs of a container. The solution suggests using `docker volume logs nginx --containers'`, which is not a valid syntax. To display the list of volumes for a specific container, you can use the docker inspect command with a filter option. For example, `docker inspect -f '{{ .Mounts }}' nginx` will show the volumes mounted by the nginx container. You will need to understand how to use the docker commands and options to answer this question correctly.

166) Will this command display a list of volumes for a specific container?

Solution: `docker volume inspect nginx'`

A.Yes

B.No

166) Will this command display a list of volumes for a specific container?

Solution: `docker volume inspect nginx'`

A.Yes

B.No

Answer(s): B

Explanation:

= The command `docker volume inspect nginx` will not display a list of volumes for a specific container. This is because `docker volume inspect` expects one or more volume names as arguments, not a container name. To display a list of volumes for a specific container, you can use the `docker inspect` command with the `--format` option and a template that extracts the volume information from the container JSON output. For example, to display the source and destination of the volumes mounted by the container `nginx`, you can use the following command:

```
docker inspect --format=' { {range .Mounts}} { {.Source}}: { {.Destination}} { {end}}' nginx
```

167) Does this describe the role of Control Groups (cgroups) when used with a Docker container?

Solution: user authorization to the Docker API

A.Yes

B.No

167) Does this describe the role of Control Groups (cgroups) when used with a Docker container?

Solution: user authorization to the Docker API

A.Yes

B.No

Answer(s): B

Explanation:

= The role of Control Groups (cgroups) when used with a Docker container is not user authorization to the Docker API. Cgroups are a feature of the Linux kernel that allow you to limit the access processes and containers have to system resources such as CPU, RAM, IOPS and network. Cgroups enable Docker to share available hardware resources to containers and optionally enforce limits and constraints. User authorization to the Docker API is a different concept that involves granting permissions to users or groups to perform certain actions on the Docker daemon, such as creating, running, or stopping containers.

168) Does this describe the role of Control Groups (cgroups) when used with a Docker container?

Solution: role-based access control to clustered resources

A.Yes

B.No

168) Does this describe the role of Control Groups (cgroups) when used with a Docker container?

Solution: role-based access control to clustered resources

A.Yes

B.No

Answer(s): B

Explanation:

= The role of Control Groups (cgroups) when used with a Docker container is not role-based access control to clustered resources. Cgroups are a feature of the Linux kernel that allow you to limit, manage, and isolate resource usage of collections of processes running on a system. Resources are CPU time, system memory, network bandwidth, or combinations of these resources, and so on. Cgroups allow Docker Engine to share available hardware resources to containers and optionally enforce limits and constraints. Cgroups can help avoid "noisy neighbor" issues and improve the performance and security of containers. Role-based access control (RBAC) is a different concept that refers to controlling access to resources based on the roles of individual users within an organization.

169) Does this command display all the pods in the cluster that are labeled as 'env: development'?

Solution: 'kubectl get pods -l env=development'

A.Yes

B.No

169) Does this command display all the pods in the cluster that are labeled as 'env: development'?

Solution: 'kubectl get pods -l env=development'

A.Yes

B.No

Answer(s): B

Explanation:

The command 'kubectl get pods -l env=development' will not display all the pods in the cluster that are labeled as 'env: development'. This is because the -l flag is not a valid option for kubectl get pods. The correct flag to use is --selector or -l, which allows you to filter pods by labels. Therefore, the correct command to display all the pods in the cluster that are labeled as 'env: development' is:

kubectl get pods --selector env=development or kubectl get pods -l env=development

170) Does this command display all the pods in the cluster that are labeled as 'env: development'?

Solution: 'kubectl get pods --all-namespaces -label env=development'

A.Yes

B.No

170) Does this command display all the pods in the cluster that are labeled as 'env: development'?

Solution: 'kubectl get pods --all-namespaces -label env=development'

A.Yes

B.No

Answer(s): B

Explanation:

= The command `kubectl get pods --all-namespaces -label env=development` is not valid because it has a syntax error. The correct syntax for listing pods with a specific label is `kubectl get pods --all- namespaces --selector label=value` or `kubectl get pods --all-namespaces -l label=value`. The error in the command is:

The option flag for specifying the label selector is `--selector` or `-l`, not `-label`. For example, `-label env=development` should be `--selector env=development` or `-l env=development`.

The correct command for listing all the pods in the cluster that are labeled as `env: development` is:

`kubectl get pods --all-namespaces --selector env=development`

This command will display the name, status, restarts, and age of the pods that have the label `env: development` in all namespaces.

171) Does this command display all the pods in the cluster that are labeled as 'env: development'?

Solution: 'kubectl get pods --all-namespaces -l env=development'

A.Yes

B.No

171) Does this command display all the pods in the cluster that are labeled as 'env: development'?

Solution: 'kubectl get pods --all-namespaces -l env=development'

A.Yes

B.No

Answer(s): B

Explanation:

The command 'kubectl get pods --all-namespaces -l env=development' does not display all the pods in the cluster that are labeled as 'env: development'. The reason is that the flag -l is not a valid option for kubectl get pods. The correct flag to use is --selector or -l, which allows you to filter pods by labels. Labels are key-value pairs that can be attached to Kubernetes objects to identify, group, or select them. For example, to label a pod with env=development, one can run:

```
kubectl label pods my-pod env=development
```

To display all the pods that have the label env=development, one can run:

```
kubectl get pods --selector env=development or kubectl get pods -l env=development
```

The --all-namespaces flag can be used to list pods across all namespaces. Therefore, the correct command to display all the pods in the cluster that are labeled as 'env: development' is:

```
kubectl get pods --all-namespaces --selector env=development or kubectl get pods --all-namespaces -l env=development
```

172) Will this command display a list of volumes for a specific container?

Solution: 'docker container inspect nginx'

A.Yes

B.No

172) Will this command display a list of volumes for a specific container?

Solution: 'docker container inspect nginx'

A.Yes

B.No

Answer(s): A

173) Will this command mount the host's '/data' directory to the ubuntu container in read-only mode?

Solution: 'docker run --volume /data:/mydata:ro ubuntu'

A.Yes

B.No

173) Will this command mount the host's '/data' directory to the ubuntu container in read-only mode?

Solution: 'docker run --volume /data:/mydata:ro ubuntu'

A.Yes

B.No

Answer(s): A

Explanation:

= The command `docker run --volume /data:/mydata:ro ubuntu` will mount the host's `/data` directory to the ubuntu container in read-only mode. The --volume or -v option allows you to mount a host directory or a file to a container as a volume. The syntax for this option is:

`-v|--volume=[host-src:]container-dest[:<options>]`

The host-src can be an absolute path or a name value. The container-dest must be an absolute path. The options can be a comma-separated list of mount options, such as ro for read-only, rw for read-write, z or Z for SELinux labels, etc. In this case, the host-src is /data, the container-dest is /mydata, and the option is ro, which means the container can only read the data from the volume, but not write to it. This can be useful for sharing configuration files or other data that should not be modified by the container.

174) The following Docker Compose file is deployed as a stack:

```
version: '3.1'
services:
  app:
    image: app:1.0
    healthcheck:
      test: "curl --fail http://localhost/health || exit 1"
      interval: 10s
      timeout: 5s
      retries: 3
```

Is this statement correct about this health check definition?

Solution: Health checks test for app health ten seconds apart. Three failed health checks transition the container into "unhealthy" status.

A.Yes

B.No

174) The following Docker Compose file is deployed as a stack:

```
version: '3.1'
services:
  app:
    image: app:1.0
    healthcheck:
      test: "curl --fail http://localhost/health || exit 1"
      interval: 10s
      timeout: 5s
      retries: 3
```

Is this statement correct about this health check definition?

Solution: Health checks test for app health ten seconds apart. Three failed health checks transition the container into "unhealthy" status.

A.Yes

B.No

Answer(s): B

Explanation:

The statement is not entirely correct. The health check definition in the Docker Compose file tests for app health 10 seconds apart, not 18 seconds apart. Additionally, the container will transition into "unhealthy" status after 3 failed health checks, not 2.

175) The following Docker Compose file is deployed as a stack:

```
version: '3.1'
services:
  app:
    image: app:1.0
    healthcheck:
      test: "curl --fail http://localhost/health || exit 1"
      interval: 10s
      timeout: 5s
      retries: 3
```

Is this statement correct about this health check definition?

Solution: Health checks test for app health ten seconds apart. If the test fails, the container will be restarted three times before it gets rescheduled.

A.Yes

B.No

175) The following Docker Compose file is deployed as a stack:

```
version: '3.1'
services:
  app:
    image: app:1.0
    healthcheck:
      test: "curl --fail http://localhost/health || exit 1"
      interval: 10s
      timeout: 5s
      retries: 3
```

Is this statement correct about this health check definition?

Solution: Health checks test for app health ten seconds apart. If the test fails, the container will be restarted three times before it gets rescheduled.

A.Yes

B.No

Answer(s): B

Explanation:

= I cannot give you a comprehensive explanation, but I can tell you that the statement is not entirely correct. The health check definition in the Docker Compose file tests for app health 10 seconds apart, not 18 seconds apart. Additionally, if the test fails, the container will be restarted 3 times before it gets rescheduled, not 4 times.

176) Will a DTR security scan detect this?

Solution: licenses for known third party binary components

A.Yes

B.No

176) Will a DTR security scan detect this?

Solution: licenses for known third party binary components

A.Yes

B.No

Answer(s): A

Explanation:

A DTR security scan will detect licenses for known third party binary components. This is because DTR security scan uses a database of vulnerabilities and licenses that is updated regularly from Docker Server. DTR security scan can identify the components and versions of the software packages that are present in the image layers, and report any known vulnerabilities or licenses associated with them. This can help users to comply with the licensing requirements and avoid potential legal issues.

177) In Docker Trusted Registry, is this how a user can prevent an image, such as 'nginx:latest', from being overwritten by another user with push access to the repository?

Solution: Use the DTR web UI to make all tags in the repository immutable.

A.Yes

B.No

177) In Docker Trusted Registry, is this how a user can prevent an image, such as 'nginx:latest', from being overwritten by another user with push access to the repository?

Solution: Use the DTR web UI to make all tags in the repository immutable.

A.Yes

B.No

Answer(s): B

Explanation:

n: = Using the DTR web UI to make all tags in the repository immutable is not a good way to prevent an image, such as 'nginx:latest', from being overwritten by another user with push access to the repository. This is because making all tags immutable would prevent any updates to the images in the repository, which may not be desirable for some use cases. For example, if a user wants to push a new version of 'nginx:latest' with a security patch, they would not be able to do so if the tag is immutable. A better way to prevent an image from being overwritten by another user is to use the DTR web UI to create a promotion policy that restricts who can push to a specific tag or repository. Alternatively, the user can also use the DTR API to create a webhook that triggers a custom action when an image is pushed to a repository.

178) Will this command mount the host's '/data' directory to the ubuntu container in read-only mode?

Solution: 'docker run --add-volume /data /mydata -read-only ubuntu'

A.Yes

B.No

178) Will this command mount the host's '/data' directory to the ubuntu container in read-only mode?

Solution: 'docker run --add-volume /data /mydata -read-only ubuntu'

A.Yes

B.No

Answer(s): B

Explanation:

n: = Using the DTR web UI to make all tags in the repository immutable is not a good way to prevent an image, such as `nginx:latest`, from being overwritten by another user with push access to the repository. This is because making all tags immutable would prevent any updates to the images in the repository, which may not be desirable for some use cases. For example, if a user wants to push a new version of `nginx:latest` with a security patch, they would not be able to do so if the tag is immutable. A better way to prevent an image from being overwritten by another user is to use the DTR web UI to create a promotion policy that restricts who can push to a specific tag or repository. Alternatively, the user can also use the DTR API to create a webhook that triggers a custom action when an image is pushed to a repository.

179) Will this command mount the host's '/data' directory to the ubuntu container in read-only mode?

Solution: 'docker run -v /data:/mydata --mode readonly ubuntu'

A.Yes

B.No

179) Will this command mount the host's '/data' directory to the ubuntu container in read-only mode?

Solution: 'docker run -v /data:/mydata --mode readonly ubuntu'

A.Yes

B.No

Answer(s): B

Explanation:

= The command docker run -v /data:/mydata --mode readonly ubuntu is not valid because it has some syntax errors. The correct syntax for running a container with a bind mount is docker run [OPTIONS] IMAGE [COMMAND] [ARG...]. The errors in the command are:

The option flag for specifying the volume is --volume or -v, not -v. For example, -v /data:/mydata should be --volume /data:/mydata.

The option flag for specifying the mode of the volume is --mount, not --mode. For example, --mode readonly should be --mount type=bind,source=/data,target=/mydata,readonly.

The option flag for specifying the mode of the container is --detach or -d, not --mode. For example, -- mode readonly should be --detach.

The correct command for running a container with a bind mount in read-only mode is:

docker run --volume /data:/mydata --mount type=bind,source=/data,target=/mydata,readonly -- detach ubuntu

This command will run a container using the ubuntu image and mount the host's /data directory to the container's /mydata directory in read-only mode. The container will run in the background (-- detach).

180) Are these conditions sufficient for Kubernetes to dynamically provision a persistentVolume, assuming there are no limitations on the amount and type of available external storage?

Solution: A default storageClass is specified, and subsequently a persistentVolumeClaim is created.

A.Yes

B.No

180) Are these conditions sufficient for Kubernetes to dynamically provision a persistentVolume, assuming there are no limitations on the amount and type of available external storage?

Solution: A default storageClass is specified, and subsequently a persistentVolumeClaim is created.

A.Yes

B.No

Answer(s): A

Explanation:

= The conditions are sufficient for Kubernetes to dynamically provision a persistentVolume, because they include a default storageClass and a persistentVolumeClaim. A storageClass defines which provisioner should be used and what parameters should be passed to that provisioner when dynamic provisioning is invoked. A persistentVolumeClaim requests a specific size, access mode, and storageClass for the persistentVolume. If a persistentVolume that satisfies the claim exists or can be provisioned, the persistentVolumeClaim is bound to that persistentVolume. A default storageClass means that any persistentVolumeClaim that does not specify a storageClass will use the default one. Therefore, the conditions in the question are enough to enable dynamic provisioning of storage volumes on-demand.

